The two-moment certificate is robust under Rudnick–Sarnak-range cubic augmentation with capacity control

An exact absorption certificate for the two-moment system at bandwidth one, with a hand-checkable witness

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Abstract

Alpöge and Furman prove unconditionally that more than two thirds of the zeros of the Riemann zeta function are simple and lie on the critical line [AF26]. Their argument rests on a two-moment certificate — the trace and the Frobenius norm of the Weil–Gabor compression, read against block structure and integrality — and that certificate is provably exhausted: within the data class (tr, Frobenius, positive-index count, integer atoms) an on-line double and a shallow off-line pair are indistinguishable, which pins the distinct-zeros optimum at $N_d/N = 5/6$ and the simple fraction at $2/3$. A natural proposal to break this degeneracy is to adjoin, at a second bandwidth $\lambda' = 1/2 + o(1)$ inside the Rudnick–Sarnak range, the unconditionally computable signed third trace together with two spectral-capacity rows: an all-V eigenvalue-count ladder and a garnish-capacity fuzz row.

This paper proves, at the model level, and certifies computationally, that the augmentation adds exactly nothing. The marginal value of the entire cubic block over the two-moment baseline is $\delta_0 = 0$ identically, an exact LP equality with an explicit, hand-checkable witness law. It holds over the full adversary grid for the distinct-zeros benchmark (940 records, $|\delta_0| < 10^{-12}$ everywhere, corner $5/6$); for the simple-fraction benchmark (corner $2/3$) at its two decision cells together with cross-checks, converged clean at the stopping criterion; and inside the near-CUE data class licensed by the unconditional pair-correlation theorem of Baluyot–Goldston–Suriajaya–Turnage-Butterbaugh [BGSTB24], where the absorption is by strict slack: there the cubic row does not bind.

The structural explanation is a set of exact identities proved here. On isolated blocks the signed cubic row is the affine function $3F - 2$ of the Frobenius row plus the second integrality level, hence blind to doubles-versus-pairs at every depth. On the ψ_1-zero grid the bandwidth-one Frobenius row degenerates exactly to the multiplicity count while the half-bandwidth kernel stays position-sensitive, so the cubic budget is met by arrangement at zero cost. Spectral escape is capped sharply at $2\sqrt{2}\sqrt{C_{\mathrm{led}}\varepsilon_g}\,N = o(N)$, and a scalar cubic tail row with any divergent cutoff is unconditionally vacuous.

The pair-interference channel, the one gap in the atom-only model theorem, is closed by an integrality theorem throughout the admissible pair-depth family ($w \le 1$, plus deep probes at $w \in \{1.5, 2\}$): proved unconditionally for single-pair columns to $w \le 2.8$, which covers

*hello.jay.tyagi@gmail.com. *Use of AI.* The mathematics in this paper — the derivations, the computations, the verification suite, the Lean formalization, and the text — was produced by Claude (Anthropic) working under the author's direction; the author set the objectives, made the technical and editorial decisions, and is responsible for the content. Claude is a tool and is not an author. The complete working record is public: https://github.com/x67ai/riemann-rh-program.

that whole family, and for multi-pair columns at $w \le 0.82$ ($w \le 0.98$ modulo one crowding cap whose ledger constant is interval-certified; on the deepest sliver $w \in (0.98, 1]$ the coverage is adversarial-numerical with the unconditional 8/9 backstop), with an unconditional 8/9-strength spectral backstop at all depths. Mixed-depth multi-pair columns involving the deep probes $w \in \{1.5, 2\}$ rest on that backstop together with converged adversarial pricing. The two-moment certificate, and with it the bandwidth-one ceiling, is robust under Rudnick–Sarnak-range cubic augmentation with capacity control.

1 Introduction

1.1 The two-moment certificate and its exhaustion

Alpöge and Furman prove unconditionally [AF26] that more than two thirds of the zeros of the Riemann zeta function are simple and on the critical line, that $N_d(T, 2T) \ge (5/6 - o(1))N(T, 2T)$ zeros are distinct, and (Theorem D, window-optimized) that the simple-on-line proportion reaches $2 - 1/c_1^* = 0.6725007\ldots$, by compressing the Weil explicit-formula form to a critically sampled Gabor family and reading exactly two spectral moments of the resulting Gram matrix $\widehat{G}$: the trace $\operatorname{tr}\widehat{G} = N(1 + o(1))$ and the Frobenius moment $\operatorname{tr}\widehat{G}^2 = (1/\lambda + \lambda/3)N(1 + o(1))$, both evaluated unconditionally on the prime side for bandwidth $\lambda \le 1$. The certificate is finite-dimensional linear algebra (the rank–trace inequality "Lemma R") applied against the zero side's block structure: on-line points give PSD rank-one atoms, off-line pairs give signature-$(1,1)$ hyperbolic blocks.

Two formalized results delimit this method exactly. First, `lemmaR_tight` (Lean, `Zeta23/ZeroSide/TightMult.lean` in the formalization accompanying [AF26]): on configurations of integer-marked on-line atoms plus pair blocks, the two-moment certificate is achieved with equality, and an on-line double (eigenvalue $+2$, charge 4) is indistinguishable from an off-line pair of depth $\to 0$ (eigenvalues ± 2, charge 4) — no further inequality in $(\mathrm{tr}, \mathrm{Frobenius}, n_+, \text{integer atoms})$ can improve it. The extremal configuration is 2/3 simple zeros plus 1/6 on-line doubles: $N_d/N = 5/6$, simple fraction 2/3. Second, the bandwidth-one ceiling (Lean, `Zeta23/PairCeiling/`): no certificate reading only bandwidth-one form-factor data certifies a simple-zero proportion above 0.6818287 (up to the recorded stability terms), witnessed by an exact-rational 256-periodic law; and any trace/Frobenius certificate is capped below $0.8453 = 2 - 2/\sqrt{3}$ at any Dirichlet-polynomial length, since $\kappa(\lambda) = 1/\lambda + \lambda/3 \ge 2/\sqrt{3}$ for all $\lambda > 0$.

1.2 The cubic proposal

Section 7.3 of [AF26] records the conditional template: under RH the third trace $\operatorname{tr}\widehat{G}^3$ is available (Hejhal [Hej94], Rudnick–Sarnak [RS96]) and the certificate can be run with the cubic weight $\omega(m) = m/2 + (2m^2 - m^3)/18 + (4/9)\mathbf{1}_{m=1}$, giving $N_d/N \ge 0.85082$ with the window $\cos(8s/5)$ (consuming also [BHB13]'s $N^s/N \ge 19/27$ on RH). The proposal examined here is that a slice of this becomes unconditional: at bandwidth $\lambda' = 1/2 + \vartheta \log\log T/\log T$, for a fixed offset exponent $\vartheta > 0$ (strictly inside the $k = 3$ Rudnick–Sarnak range $X^3 \le T^{2-\varepsilon}$), the signed cubic trace is computable by the diagonal method of [AF26, §5], and a Schatten-3 spectral-tail theorem would cap the spectral escape that the odd-moment no-go of [AF26] (Section 7.5(e) of the 35-page version; the same item is §7.2(e) of v5, the two versions being distinguished in the References) identifies, making the cubic row a usable sign-sensitive constraint. That spectral-tail theorem has two forms, and this paper treats both: the divergent-cutoff form in which the proposal stated it — a single scalar tail row $\sum_{|\lambda_i| \ge V_0} |\lambda_i|^3 = o(N)$ above one cutoff $V_0 = V_0(T) \to \infty$ — and the stronger all-V eigenvalue-count ladder $n(V) \le C_{\mathrm{led}} N V^{-4}$, holding at every threshold rather than above a single cutoff ($\vartheta < 1$ strict). The same Chebyshev argument yields the ladder, but only through a step —

from a bound above one cutoff to a bound at every threshold — that is not proved (Section 8.3). Theorems 7.4 and 7.1 treat the two forms in turn. The proposal's separating premise was that a hyperbolic pair block contributes $(+a)^3 + (-a)^3 = 0$ to the cubic while a double contributes $+8$, so the two-moment-degenerate configurations would sit at different corners of the augmented polytope.

Item (e) of [AF26] is not silent on higher moments. In the 35-page version it reads, verbatim:

> "(e) The prime-side evaluation of $\operatorname{tr}\widetilde{G}^k$ by the diagonal method of Section 5 (multiplicative relations among k prime powers, Montgomery–Vaughan for the rest) is available exactly in the Rudnick–Sarnak range $k\lambda < 2$ [RS96]; for $\lambda \in (1/2, 1)$ this allows at most $k = 3$ (and only for $\lambda < 2/3$), and an odd moment does not lower $\Lambda_1(0)$. Thus, unconditionally, higher moments add nothing to the n_+-bound on $(1/2, 1)$ (and Proposition 4.4 uses only the first two in any case), while for $\lambda \le 1/2$, where many moments are available, Proposition 7.4 makes them useless."

(In v5 the same item is compressed to "… at $X \asymp T$ this allows only $k = 1$. Thus, unconditionally, higher moments add nothing.") The remark is a single-bandwidth statement, and it disposes of the single-bandwidth question twice over, for two disjoint reasons: at bandwidth $\lambda \in (1/2, 1)$ an odd moment does not move the Christoffel bound, and at $\lambda \le 1/2$, where moments are plentiful, the certificate is worthless on its own (its Proposition 7.4 cap is non-positive there). Alpöge and Furman themselves label the remark informal.

The question addressed here is a two-bandwidth one, which that remark does not reach: keep the bandwidth-one Frobenius baseline and adjoin a second-bandwidth block at $\lambda' = 1/2$ — the λ'-Frobenius budget, the signed cubic equality, the all-V count ladder, and the capacity fuzz row (the widening of the cubic equality that Frobenius slack can purchase; rows (R-6) and (R-5′) of Section 2.5). Rows that are useless alone can bind when adjoined to a strong baseline, so what the two-bandwidth question needs is a marginal-value computation rather than a restatement; and the remark addresses neither the distinct-zeros benchmark, nor the pair-block structure, nor the near-CUE data class, and exhibits no witness. The answer obtained here is $\delta_0 = 0$ as an exact LP equality, witness-backed, on both benchmarks, and by strict slack inside the near-CUE class, together with the structural identities (Sections 3–4) and the capacity theorem (Section 7) that explain it.

Related work. Gonçalves, de Laat and Leijenhorst [GdLL25] run the higher-level-correlation program against multiplicity directly, by semidefinite programming over an auxiliary-function class: on GRH they obtain, from 3-level correlations, that at least 96.14% of the zeros of a primitive L-function have multiplicity at most 2, and their $n = 3$ row pays where their $n = 2$ row does not. That is consistent with the result proved here: theirs is a single-bandwidth, conditional statement bounding *multiplicity* at their support parameter, whereas the one proved here is an unconditional statement about the *marginal value of adjoining a second bandwidth* to a fixed bandwidth-one Frobenius baseline, at effective support 1 (their $m = 2$ row), measured against a fixed objective. The combinatorial half of their apparatus and of this paper's is the same identity — their equation (5) is the Stirling change of basis between the power sums $N_n = \sum_{\text{zeros}} m^n$ and the falling-factorial counts $M_k = \sum_{\text{zeros}} m(m-1)\cdots(m-k+1)$, whose $n = 3$ case is the atom half of Theorem 3.4 below — so the atom half is classical, and only the pair term $6\sum_p m_p^2(m_p - 1)A_p^2$ is new. The remark following their (6) records that adapting their optimization to bound N_n directly did not improve on the bounds obtained through (5); that is a negative about problem *formulation* rather than about the information content of a higher correlation level. A closer structural comparandum is sphere packing, where the three-point bound of Cohn, de Laat and Salmon [CdLS22] *does* improve on the two-point Cohn–Elkies bound in several dimensions: third-order augmentation buying exactly nothing here is therefore a substantive negative, not a foregone one.

The marginal-value question is settled here by a linear program over an adversary class — marks up to a divergent alphabet W, depth-parametrized shallow pairs, clustered sine-process null budgets, full positional freedom, and an explicit garnish-capacity fuzz row derived from the all-V ladder. A strictly positive marginal value would be a gain to be developed analytically; a marginal value of zero — an *absorption* of the cubic block by the two-moment optimum — sharpens the no-go. The value is zero.

1.3 The marginal-value question and its answer

Write P_{base} for the minimum of $E_w[N_d]/N$ over laws w (probability mixtures) of explicit N-periodic marked configurations satisfying the two-moment data — mass N, integrality, pair structure, bandwidth-one Frobenius budget $(4/3)N(1 \mp \varepsilon)$ — and P_{full} for the same minimum after the entire $\lambda' = 1/2$ cubic block is adjoined (the λ'-Frobenius budget $\kappa(1/2)N = (13/6)N$, the signed cubic equality $c_3(1/2)N = 5N$, the all-V count ladder, and the capacity fuzz row). Both are linear programs, and Section 2 fixes every term. The question is whether the cubic block has strictly positive marginal value, $\delta_0 := P_{\text{full}} - P_{\text{base}} > 0$.

Throughout, "Rudnick–Sarnak-range cubic augmentation with capacity control" (the title phrase) means exactly this $\lambda' = 1/2 + o(1)$ system, with $\vartheta < 1$ strict — the only point in the RS range where both the signed cubic row and the capacity ladder are unconditionally licensed; the behavior of the cubic row at $\lambda' \geq 0.6$, inside the RS range but outside the proven ladder regime, is the exploratory payoff curve of Section 8.4, not part of any claim.

The answer, certified computationally over the decision grid and explained by the model-level theorems below, is no: $\delta_0 = 0$ identically, not merely within error bars, since the base-optimal law itself satisfies the entire cubic block. The rest of the paper assembles the result: the structure theorems that explain the absorption (Section 3), the model-level corner theorem with its pair-channel extension (Section 4), the computational certification with the explicit witness (Section 4.4), the near-CUE result (Section 5), the simple-fraction benchmark (Section 6), the two capacity theorems that close the spectral-escape routes (Section 7), and the scope map of what is exact, what is a bounded residual, and what remains open (Section 8).

1.4 Headline results

1. **Exact absorption with witness (Theorem 4.3 + Section 4.4).** Over the full decision grid — 940 LP records spanning mark alphabets $W \in \{2, 3, 4, 6, 8\}$, ladder constants $C_{\text{led}} \in \{10, 100, 1000\}$, tolerances $\varepsilon \in \{0.02, 0.05, 0.10\}$ (with the separate, labeled tight-ε runs extending to $\varepsilon = 0.002$), all fuzz modes, matched and asymptotic budget centerings, $N = 64$ and $N = 128$ — the marginal value of the cubic block is $|\delta_0| < 10^{-12}$ at every record (maximum 8.94×10^{-13}), and the joint optimum obeys the exact law $P(\varepsilon) = 5/6 - (2/3)\varepsilon \to 5/6$. The absorbing law is exhibited: three columns on the 65-site ψ_1-zero grid, marks $\{1, 2\}$ only, no pairs, every row verified in exact rational arithmetic, the key entries hand-checkable ($F_1 = 64$ exactly; $F' = 4128/33$).

2. **Both benchmarks.** The same absorption holds for the simple-fraction objective: the cubic block moves the p_1 optimum by at most 1.4×10^{-12}, converged at the stopping criterion (no improving column in $S = 200$ pricing restarts, for the full and base systems alike), and the optimum is the exact analytic corner $2 - (4/3)(1 + \varepsilon) \to 2/3$ (Section 6).

3. **Near-CUE absorption by slack (Section 5).** Inside the pinned data class $\left|E_w|c_j|^2 - j\right| \le 1$ (licensed unconditionally, with three stated riders, by [BGSTB24]), the marginal value of the cubic block in that pinned class is $\delta_0' = 0$, an exact LP equality at $N = 64$ and $N = 128$, at both budget centerings — and at the pinned optimum only the pinning rows bind: the cubic row, the ladder, and even both Frobenius rows are strictly interior.

4. **Cubic blindness (Theorem 3.4).** On isolated blocks, the signed cubic row equals $3F - 2$ per zero on every multiplicity-1 pair at every depth and on every mark-$\{1,2\}$ atom: $C - 3F + 2\,\mathrm{Mass} = \sum m(m-1)(m-2) + 6\sum m_p^2(m_p - 1)A_p^2$ exactly. The "+8 vs 0" separating premise of the proposal is false: a pair's cubic charge is $2m^3(1 + 3A^2) \ge 8m^3$ at every depth. All discriminating power of the cubic row lives in marks ≥ 3 and in clustering cross-terms — and the cross-terms are a free resource for the adversary (Theorem 3.9).

5. **Closed forms and the $m_2 = \kappa$ identity (Theorem 3.5).** The sine-process null budgets are $m_2(\lambda) = 1/\lambda + \lambda/3$ and $m_3(\lambda) = 1 + 1/\lambda^2$ for the flat window, all $0 < \lambda \le 1$; and $m_2(\lambda)$ coincides exactly with the unconditional prime-side $\kappa(\lambda)$ of [AF26]: the sine null's second-moment budget is the unconditional budget, on the whole band.

6. **Capacity cap (Theorem 7.1).** Under the all-V ladder and a Frobenius slack ε_g, the maximal cubic charge purchasable by spectral escape is exactly $2\sqrt{2}\sqrt{C_{\mathrm{led}}\varepsilon_g}\,N = o(N)$ — sharp, with the extremal profile exhibited. The constant is confirmed by three independent derivations, and it supersedes two earlier estimates, the two-term dyadic $2\sqrt{C\varepsilon}$ and the point-mass profile's $(5/\sqrt{3})\sqrt{C\varepsilon}$.

7. **Divergent-cutoff vacuity (Theorem 7.4).** A scalar Schatten-3 tail row with any divergent cutoff is unconditionally absorbed by an explicit "garnish" construction — isolated on-line atoms placed below the tail row's cutoff — that lowers both objectives; this half of the no-go needs no linear program at all.

8. **Pair-interference closure (Sections 4.2–4.3).** The one channel outside the atom-only sandwich — off-line pairs pushing the bandwidth-one Frobenius below the mark-accounting floor — is closed for the admissible pair class (Section 2.4) throughout its depth family: proved unconditionally for all single-pair configurations to depth $w \le 2.8$ (covering every admissible depth, including both deep probes, with $\ge 34\%$ margin), for all multi-pair configurations to $w \le 0.82$ ($w \le 0.98$ modulo the crowding cap, ledger constant interval-certified at 0.98465; on $w \in (0.98, 1]$ the ledger fails — certified — and coverage is adversarial numerics plus the 8/9 backstop), and for equal-depth pair-only configurations at every depth. Multi-pair configurations at mixed depths involving the deep probes $w \in \{1.5, 2\}$ — admissible under Section 2.4 but beyond the theorems' range — are covered by adversarial numerics plus the unconditional 8/9 backstop (see (O2), Section 8.3). An explicit fractional-mark counterexample proves the closure is intrinsically an integrality theorem.

1.5 What the cubic row reads, and the scope of the results

The cubic row is a nonlinear spectral reading of bandwidth-one pair-correlation data with quantitative high-spectrum capacity control. The third trace at bandwidth λ' carries total Fourier support $(k-1)\lambda' = 1 + o(1)$, not $3\lambda'$; its genuinely cubic prime resonance is the absolute constant $\sum_p (\log p)^3/(p-1)^2 = 2.315762$ (evaluated by sieving to 3×10^6), contributing $O(1)$ to $\operatorname{tr}\widehat{G}^3$, i.e. $O(1/N)$ per zero. No claim is made here that the cubic block evades the Alternative Hypothesis, or that it consumes data beyond the bandwidth-one class plus $o(1)$; for what the currently known band-limited n-level data provably cannot do against AH, see Lagarias–Rodgers [LR20] and Tao's independent treatment cited there. The no-go proved here is precisely that this nonlinear reading, with the full capacity apparatus attached, adds nothing to the linear reading at the proven operating point. Formally, a cubic row is outside the PairCeiling certificate class as defined; the content of this paper is that leaving the class buys nothing.

Every theorem in this paper is a statement about the model — the explicit configuration classes, kernels, and row systems of Section 2 — not about the zeros of zeta. The bridge from the model rows to unconditional statements about zeta is the sampling bookkeeping of [AF26] plus the licensing riders of [BGSTB24], stated in full in Section 5.1. The exact zero $\delta_0 = 0$ is proved and certified at the flat/flat window configuration and is specific to it: with the Montgomery–Taylor cosine window at $\lambda = 1$ the grid degeneracy breaks and the statement is a bounded residual of order 6×10^{-5} (Section 8). What is established is therefore not a universal "$\delta_0 = 0$": an exact zero on the decision grid and in the near-CUE class; bounded residuals on the window family, each below the 10^{-2} scale at which the question was posed; a proven $o(N)$ cap on spectral escape everywhere; and an exploratory payoff curve outside the proven ladder regime (Section 8.4).

2 The model

This section fixes the model in which every theorem of Sections 3–7 lives: the master trace formula, its finite-circle discretization, the adversary class, and the row system.

2.1 Window and kernel

Fix a window profile $v \geq 0$ on $[-1/2, 1/2]$, even, with $\int v > 0$ (the scale-free profile of the taper-squared ϕ^2 of Alpöge and Furman [AF26]; their formalized taper class is C^3). The *normalized sampling kernel* at bandwidth $\lambda > 0$ is the entire function

$$\psi_\lambda(x) = \frac{\displaystyle\int_{-1/2}^{1/2} v(\sigma)\, e^{2\pi i \lambda \sigma x}\, d\sigma}{\displaystyle\int v},$$

so $\psi_\lambda(0) = 1$, with Fourier transform u supported in $|\alpha| \leq \lambda/2$ (Montgomery α-units), $\int u = 1$. For the flat window $v = 1$:

$$\psi_\lambda(x) = \operatorname{sinc}(\pi\lambda x) = \frac{\sin(\pi\lambda x)}{\pi\lambda x}, \qquad u = \frac{1}{\lambda}\,\mathbf{1}_{[-\lambda/2,\,\lambda/2]}.$$

Positions x are in mean-gap units (mean zero spacing 1); complex positions (off-line zeros) are allowed, ψ_λ being entire.

2.2 Continuum model: configurations and the master trace formula

A *marked configuration* is a finite multiset of points z with positions γ_z (complex allowed) and integer marks $m_z \geq 1$; off-line points occur only in conjugate pairs $\{\gamma + id, \gamma - id\}$ with common depth and common mark (functional-equation symmetry). The *master trace formula* defines the k-th trace at bandwidth λ:

$$\operatorname{tr} G^k := \sum_{z_1,\ldots,z_k} \Big(\prod_j m_{z_j}\Big)\, \psi_\lambda(\gamma_{z_1} - \gamma_{z_2})\, \psi_\lambda(\gamma_{z_2} - \gamma_{z_3}) \cdots \psi_\lambda(\gamma_{z_k} - \gamma_{z_1}),$$

summed over ordered k-tuples of configuration points. This is the Poisson sampling identity of [AF26] in normalized units (units in which an isolated on-line zero of mark m has eigenvalue exactly m), and it is realized by an actual Hermitian matrix: Section 3.1 constructs the block matrices explicitly where spectral (not just trace) data are needed. We write $F = \operatorname{tr} G^2$ (Frobenius row) and $C = \operatorname{tr} G^3$ (signed cubic row); F_1 at bandwidth 1, F' and C' at bandwidth λ'. This single formula generates every trace row on both sides — null budgets and configuration demands — including all clustering cross-terms and all depth effects; no isolated-block idealization enters the row system.

2.3 Finite-circle model (the LP's discretization)

Period N (zeros per period) on the circle of circumference N; configurations are N-periodic, described by one period. For the flat window at bandwidth λ the harmonic set is $B = \{j \in \mathbf{Z} : |j| \leq \lambda N/2\}$, with $M = |B|$ harmonics and uniform weights $u_j = 1/M$. With the form factor

$$c_s = \sum_z m_z \, e^{-2\pi i s \theta_z / N}$$

(a conjugate pair at $\theta \pm id$ contributes $2m \cosh(2\pi s d/N)\, e^{-2\pi i s\theta/N}$), the trace rows are

$$\begin{aligned}
\operatorname{tr} \widehat{G} &= \sum_z m_z && \text{(mass)},\\
\operatorname{tr} \widehat{G}^2 &= \sum_s W_2(s)\, |c_s|^2, && W_2(s) = \sum_{j\in B,\ s-j\in B} u_j u_{s-j},\\
\operatorname{tr} \widehat{G}^3 &= \sum_{k_1,k_2} U_3(k_1,k_2)\, c_{k_1} c_{k_2} c_{-k_1-k_2}, && U_3(k_1,k_2) = \sum_j u_{j+k_1} u_{j+k_1+k_2} u_j.
\end{aligned}$$

Eigenvalue data (ladder counts, inertia) come from the Hermitian frequency matrix $H[j,j'] = \sqrt{u_j u_{j'}}\, c_{j-j'}$ $(j, j' \in B)$, whose k-th trace equals the k-th trace row exactly (alias-free tight-frame compression). At the primary geometry $N = 64$, $\lambda = 1$: $M = 65$; $\lambda' = 1/2$: $M' = 33$. In position space, for a symmetric band, $\operatorname{tr} \widehat{G}^2 = \sum_{z,z'} m_z m_{z'} D(\theta_z - \theta_{z'})^2$ with $D(x) = \sum_j u_j e^{-2\pi i j x/N}$ real, even, $D(0) = 1$ — the discrete analog of the continuum $\operatorname{tr} G^2 = \sum m\, m'\, \psi_\lambda^2$.

2.4 Adversary class, laws, and the two benchmarks

A *column*, also called an *admissible* configuration below, is an explicit N-periodic marked configuration: on-line atoms (positions $\theta_i \in [0, N)$ — positions are written θ throughout, not to be confused with the offset exponent ϑ of Section 1 — integer marks $1 \leq m_i \leq W$) and off-line pairs (position, dimensionless depth w in the admissible pair-depth family $(0, 1] \cup \{1.5, 2\}$, the last two values being the deep probes, pair multiplicity m), with mass $\sum m_i + 2 \sum m_p = N$ per period. A *law* is a probability mixture $w_c \geq 0$, $\sum_c w_c = 1$ over columns; every row is $E_w[\text{row}(c)]$. The two objectives:

$$\begin{aligned}
N_d(c) &= \#\text{atoms} + 2\#\text{pairs} && \text{(distinct zeros; only on-line multiples reduce it)},\\
p_1(c) &= \big(\#\text{mark-1 atoms} + 2\#\text{mult-1 pairs}\big)/N && \text{(simple fraction)}.
\end{aligned}$$

Realizability. Imposed: explicit finite marked multisets (never bare correlation data); integer marks; conjugate-pair structure; mass N; mixtures. *Necessity:* an adversary allowed to post aggregated moment data directly could claim values realized by no point configuration (unconstrained cross-term variables can leave the convex hull of realizable moment pairs), making an "absorption" vacuous; this is the truncated-moment realizability problem of Kuna, Lebowitz and Speer [KLS07, KLS11], of which the finite-circle argument here is a special case. Integrality, pair symmetry, and density are properties the true zero multiset provably has, and dropping any of them hands the adversary provably impossible worlds: with fractional marks even the 5/6 baseline is false, since mass N of mark-4/3 atoms on the grid of Theorem 3.9 has $F_1 = (4/3)N$ budget-tight and $N_d = (3/4)N < (5/6)N$ (Section 4.2 exhibits the subtler pair-channel form of the same failure). *Sufficiency:* nothing further may be imposed, because every row of the system is a theorem valid for arbitrary configurations with these properties — trace identities are linear algebra plus the sampling identity, the budgets are the unconditional prime-side equalities, the ladder is the all-V spectral-tail statement, inertia is Sylvester; imposing anything more (e.g. pointwise GUE 3-level correlations) would smuggle in precisely the unproven correlation knowledge whose absence the row system respects. Periodicity and the finite depth/mark grids are discretizations, refined under a stability requirement ($N = 64 \to 128$, W-scan, depth-grid halving), not constraints.

2.5 The row system

Baseline (two-moment / `lemmaR_tight` data): (R-1) mass N per period; (R-2) the bandwidth-one Frobenius budget $E_w[F_1] \in (4/3)N(1 \mp \varepsilon)$; (R-3) integrality and pair structure, imposed by construction; (R-4) the objective. In the matched-geometry variant the finite-N CUE-sampled centers B_1, B_2, B_3 replace $(4/3)N$, $\kappa(1/2)N$ and $c_3(1/2)N$ respectively; their $N = 64$ values are given in Appendix A. The cubic block at $\lambda' = 1/2$ adds:

(R-5) λ'-Frobenius: $E_w[F'] \in \kappa(1/2)N(1 \mp \varepsilon)$, $\kappa(1/2) = 13/6$;

(R-6) signed cubic with capacity coupling: $\left|E_w[C'] - c_3(1/2)N\right| \le \varepsilon_3 N + 2\sqrt{2}\sqrt{C_{\mathrm{led}} g_F}\, N$, $c_3(1/2) = 5$, the cubic tolerance being tied to the Frobenius one, $\varepsilon_3 = \varepsilon$. In the decoupled (parametric) fuzz mode the capacity term is replaced by a fixed cubic slack ΓN with $\Gamma \in \{0, 0.05, 0.10, 0.25\}$, and the shadow price $d\delta_0/d\Gamma$ is reported;

(R-5′) garnish Frobenius budget $g_F \ge 0$ charged against the Frobenius slack;

(R-7) all-V count ladder: $E_w[n(V)] \le C_{\mathrm{led}} N V^{-4}$ on the V-grid $\{2, 3, 4, 6, 8, 12, 16\}$;

(R-8) inertia: $n_-(c)$ is at most the number of pairs in c (automatic on explicit spectra; asserted);

(R-9) Frobenius split, power-mean, Hölder-with-$o(N)$: automatically satisfied by explicit spectra (on explicit configurations the linear-algebra-true rows cannot be violated, so their entire content is enforced leak-free by construction).

The budget centers are the clustered sine-process null moments (Section 3.3) — never isolated-block values — because at $\lambda' = 1/2$ the window width is two mean gaps and the null itself is cross-term-dominated: cross-terms (occupancy ≥ 2) carry 54% of the Frobenius budget and 80% of the cubic budget (analytic fractions 7/6 of 13/6 and 4 of 5; null MC 52.6% / 79.0% at $n = 64$). The decision rule: absorption iff $\delta_0 = P_{\mathrm{full}} - P_{\mathrm{base}} = 0$ within LP tolerance at the decision points with an explicit witness law; the witness must satisfy every row at centered budgets with $E[N_d]/N \le 5/6 + 0.01$. The objective, the adversary class, the rows, the decision rule and the stopping criterion were fixed in writing before the computation was run.

3 Structure theorems

On isolated blocks the signed cubic row is an affine function of the mass and Frobenius rows plus the second integrality level, so it cannot see doubles-versus-pairs at any depth (Theorem 3.4). The null budgets are then computed in closed form, yielding the identities that organize the entire margin structure, and the grid-decoupling theorem shows that the cross-term channel is available to the adversary at no cost. Every proof in this section is complete; the numerical confirmations (`results/a4-no-go/verify/verify_t1.py`–`verify_t4_mc.py`) are summarized at the point of use.

3.1 The exact pair block

Setting. Windows $\tau_k = k(2\pi/L)$, $k \in \mathbf{Z}$ (critical spacing); taper ϕ real, even, continuous, supported in $[-L/2, L/2]$, $\phi \in C^2$ (the class formalized in [AF26] is C^3 — more than enough). Fourier convention $\widehat{\phi}(z) = \int \phi(s)e^{-isz}\,ds$ (entire, by Paley–Wiener); $\Phi(z) = \int \phi(s)^2 e^{-isz}\,ds$; $a = L^{-1}\int \phi^2$, so $\Phi(0) = La$. Normalized units: matrices divided by $L^2 a$, in which an isolated on-line atom of mark m has eigenvalue m.

Lemma 3.1 (Poisson sampling identity, complex arguments)**.** *For all complex τ, τ':*

$$\sum_{k\in\mathbf{Z}} \widehat{\phi}(\tau - \tau_k)\,\widehat{\phi}(\tau' - \tau_k) = L\,\Phi(\tau - \tau')\,,$$

the sum converging absolutely and locally uniformly.

Proof. First real arguments. Fix $\tau, \tau' \in \mathbf{R}$ and set $g(t) = \widehat{\phi}(\tau - t)\,\widehat{\phi}(\tau' - t)$; $\widehat{\phi}$ is real on $\mathbf{R}$ (ϕ real even) and, since $\phi \in C^2$ with compact support, $|\widehat{\phi}(x)| \le C/(1+x^2)$, so $g \in L^1$ with $|g(t)| \le C'/(1+t^2)^2$ and the lattice sum converges absolutely. Each factor $t \mapsto \widehat{\phi}(\tau - t)$ has Fourier transform $e^{-i\tau\xi}2\pi\phi(\xi)$ (inversion), supported in $[-L/2, L/2]$; hence $\widehat{g} = (1/2\pi)$(convolution of the factor transforms) is continuous, supported in $[-L, L]$, and vanishes at the endpoints $\pm L$ (the convolution of two continuous functions supported in $[-L/2, L/2]$, evaluated at L, integrates over the null set $\{L/2\}$). Poisson summation over the lattice $(2\pi/L)\mathbf{Z}$ (classical form; $|g| + |\widehat{g}|$ summable on the respective lattices):

$$\sum_k g(2\pi k/L) = \frac{L}{2\pi}\sum_{n\in\mathbf{Z}} \widehat{g}(nL) = \frac{L}{2\pi}\,\widehat{g}(0),$$

all $n \ne 0$ terms vanishing by the support ($|nL| \ge L$, endpoint value 0). By Plancherel, $\widehat{g}(0) = \int \widehat{\phi}(\tau - t)\widehat{\phi}(\tau' - t)\,dt = (1/2\pi)\int (2\pi\phi(\xi))^2 e^{-i(\tau-\tau')\xi}\,d\xi = 2\pi\,\Phi(\tau - \tau')$, giving the identity on $\mathbf{R}^2$. For complex arguments: $|\widehat{\phi}(x+iy)| \le e^{L|y|/2}C/(1+x^2)$, so on compact subsets of $\mathbf{C}^2$ the lattice sum converges absolutely and uniformly and is entire in (τ, τ'), as is $L\Phi(\tau - \tau')$; the two entire functions agree on $\mathbf{R}^2$, hence everywhere (identity theorem, one variable at a time). $\square$

Proposition 3.2 (exact off-line pair block — the true Prop-4.1 spectrum)**.** *Let the conjugate pair be $\{\gamma + i\delta, \gamma - i\delta\}$, $\gamma \in \mathbf{R}$, depth $\delta > 0$, multiplicity m, with evaluation vector $x = (x_k)$, $x_k = \widehat{\phi}(\gamma + i\delta - \tau_k)$, and block matrix $M = m\,(xx^{\mathsf{T}} + \bar{x}\bar{x}^{\mathsf{T}})$ (transpose, not conjugate-transpose: the two rank-one terms are the pair's two zeros). Then, exactly:*

(i) $x^{\mathsf{T}}x = L\,\Phi(0) = L^2 a$ — real, independent of depth and position;

(ii) $|x|^2 = L\,\Phi(2i\delta) = L^2 a\,A(\delta)$, $A(\delta) = \int \phi^2(s)\cosh(2\delta s)\,ds/\int \phi^2(s)\,ds \ge 1$;

(iii) writing $x = p + iq$: $\langle p, q\rangle = 0$ exactly, $|p|^2 = L^2 a(1+A)/2$, $|q|^2 = L^2 a(A-1)/2$;

(iv) $M = 2m\,(pp^{\mathsf T} - qq^{\mathsf T})$*, rank* ≤ 2 *real symmetric, with nonzero eigenvalues (normalized by* L^2a*)*

$$m\,(1 + A(\delta)) \quad \textit{and} \quad m\,(1 - A(\delta)),$$

independent of the position γ*;*

(v) normalized charges: trace $2m$*; Frobenius* $2m^2(1 + A^2)$*; signed cubic* $2m^3(1 + 3A^2)$*.*

In dimensionless depth $w := L\delta$*,* A *depends on the pair only through* w*, with* $A(0+) = 1$ *and* A *strictly increasing; for the flat taper* $A(w) = \sinh(w)/w$*, with values* $A(1/8) = 1.002606$*,* $A(1/4) = 1.010449$*,* $A(1/2) = 1.042191$*,* $A(3/4) = 1.096422$*,* $A(1) = 1.175201$*,* $A(2) = 1.813430$*.*

Proof. (i) Lemma 3.1 at $\tau = \tau' = \gamma + i\delta$: $\sum_k \widehat{\phi}(\gamma + i\delta - \tau_k)^2 = L\Phi(0) = L\int \phi^2 = L^2a$ — real, independent of γ and δ. (ii) ϕ real and even gives $\widehat{\phi}(\bar z) = \overline{\widehat{\phi}(z)}$; hence $|x_k|^2 = \widehat{\phi}(u - \tau_k)\,\widehat{\phi}(\bar u - \tau_k)$ with $u = \gamma + i\delta$, and Lemma 3.1 at $(u, \bar u)$ gives $|x|^2 = L\Phi(2i\delta) = L\int \phi^2(s)\cosh(2\delta s)\,ds$ (ϕ^2 even); $A \ge 1$ by $\cosh \ge 1$, strictly increasing by strict monotonicity of cosh on the support. (iii) $x^{\mathsf T}x = |p|^2 - |q|^2 + 2i\langle p, q\rangle$; by (i) it is real, so $\langle p, q\rangle = 0$ and $|p|^2 - |q|^2 = L^2a$; with $|p|^2 + |q|^2 = |x|^2 = L^2aA$, solve. (iv) $xx^{\mathsf T} + \bar x\bar x^{\mathsf T} = (p + iq)(p + iq)^{\mathsf T} + (p - iq)(p - iq)^{\mathsf T} = 2(pp^{\mathsf T} - qq^{\mathsf T})$; with $p \perp q$ the eigenvectors are p, q with eigenvalues $2m|p|^2 = mL^2a(1 + A)$ and $-2m|q|^2 = mL^2a(1 - A)$; all others 0; normalize. Position-independence: the eigenvalues depend only on $|p|^2, |q|^2$, which depend only on δ. (v) Trace $m(1 + A) + m(1 - A) = 2m$; Frobenius $m^2(1 + A)^2 + m^2(1 - A)^2 = 2m^2(1 + A^2)$; cubic $m^3(1 + A)^3 + m^3(1 - A)^3 = 2m^3(1 + 3A^2)$. Flat-taper A: $A(w) = \int_{-1/2}^{1/2} \cosh(2w\sigma)\,d\sigma = \sinh(w)/w$; the listed values are its evaluations (they reproduce the tabulated values used in the computation to their displayed precision, $\le 4 \times 10^{-6}$). □

Consistency and inertia. The two-point configuration $\{\gamma \pm i\delta\}$ under the master formula has $\operatorname{tr} G^k = m^k \operatorname{tr} K^k$ with the 2×2 kernel $K = \left[\begin{smallmatrix} 1 & A \\ A & 1 \end{smallmatrix}\right]$, eigenvalues $1 \pm A$ — the same charges. The matrix route (iv) additionally certifies the inertia (one positive, one negative eigenvalue for $\delta > 0$) consumed by the n_- and ladder rows.

Block continuity, exact form. As $\delta \to 0+$, the spectrum tends to $(2m, -0)$, Frobenius to $2m^2$, cubic to $8m^3$: a shallow pair is row-indistinguishable from an on-line double in every row, continuously. Moreover the cubic charge $2m^3(1 + 3A^2) \ge 8m^3$ at every depth ($A \ge 1$), increasing with depth: a deep pair never loses cubic charge. The "+8 vs 0" separation premise is false under the true block law: it is the cubic bookkeeping that fails, the block spectrum $mL^2a(1 \pm A)$ of Proposition 3.2 being unaffected.

Numerical confirmation (`verify_t3.py`): evaluation-vector route with a cos taper, $L = 40$, 120001 windows: $|x^{\mathsf T}x/L^2a - 1| \le 6.7 \times 10^{-16}$, $|\langle p, q\rangle|/L^2a \le 1.7 \times 10^{-14}$, block eigenvalues vs $m(1 \pm A)$ to 7.8×10^{-14}, position-independent at machine precision at every depth $w \in \{1/64, 1/8, 1/4, 1/2, 3/4, 1, 2\}$; shallow-pair cubic charge 8.0005 at $w = 1/64$ and 8.0313 at $w = 1/8$, matching the independently computed 8.0005/8.033 — never 0 at any depth.

3.2 The affine-plane identity and cubic blindness

Definition (per-zero charges)**.** For an isolated block of total mass μ ($\mu = m$ for an atom of mark m; $\mu = 2m$ for a pair of multiplicity m), let $F = $ (block Frobenius)$/\mu$ and $c = $ (block cubic)$/\mu$. By Proposition 3.2(v) and the atom spectrum:

$$\begin{aligned} \text{atom, mark } m: &\quad F = m, && c = m^2; \\ \text{pair, mult } m, \text{ depth}: &\quad F = m(1 + A^2), && c = m^2(1 + 3A^2). \end{aligned}$$

Proposition 3.3 (affine-plane identity).

$$\begin{aligned} \textit{atom of mark } m: &\quad c = 3F - 2 + (m-1)(m-2)\,; \\ \textit{pair of mult } 1,\ \textit{any depth}: &\quad c = 3F - 2 \quad \textit{exactly}\,; \\ \textit{pair of mult } m,\ \textit{any depth}: &\quad c = 3F - 2 + (m-1)(m-2) + 3m(m-1)A^2\,. \end{aligned}$$

In particular every mark-$\{1,2\}$ atom and every multiplicity-1 pair — at every depth — lies on the same affine plane $c = 3F - 2$ in the per-zero (F, c) plane.

The atom half in the literature. For atoms the identity $c - 3F + 2 = (m-1)(m-2)$, and its summed form in Theorem 3.4, is the third-order Stirling change of basis between power sums and falling factorials: $m(m-1)(m-2) = m^3 - 3m^2 + 2m$, the coefficients being the signed Stirling numbers of the first kind $s(3,3) = 1$, $s(3,2) = -3$, $s(3,1) = 2$. The inverse conversion, $\sum m^n = \sum_k S(n,k)M_k$ with Stirling numbers of the second kind, is equation (5) of [GdLL25], where it plays the same role — including the same consequence, that the falling-factorial term vanishes identically once every multiplicity is at most $n-1$. The atom half is thus classical; what Theorem 3.4 adds is the pair term $6\sum_{\text{pairs}} m_p^2(m_p-1)A_p^2$, a statement about the finite-circle block law of Proposition 3.2 with no counterpart in the multiplicity-statistics literature.

Proof. Atom: $3F - 2 + (m-1)(m-2) = 3m - 2 + m^2 - 3m + 2 = m^2 = c$. Pair: $c - 3F + 2 = m^2(1+3A^2) - 3m(1+A^2) + 2 = (m^2 - 3m + 2) + 3A^2(m^2 - m) = (m-1)(m-2) + 3m(m-1)A^2$. At $m = 1$ both terms vanish identically in A (every depth); at $m = 2$ the atom identity also reads $c = 3F - 2$ (doubles are on the plane). $\square$

Theorem 3.4 (cubic blindness on isolated blocks). *Let the configuration be a disjoint union of isolated blocks (atoms of marks m_i; pairs of multiplicities m_p at depths with $A_p = A(\delta_p)$), with total mass* Mass*, Frobenius F, signed cubic C (sums of block charges). Then, exactly:*

$$C - 3F + 2\,\mathrm{Mass} = \sum_{\text{zeros}} m(m-1)(m-2) + 6\sum_{\text{pairs}} m_p^2(m_p-1)A_p^2\,,$$

where the first sum runs over the zero multiset (an atom of mark m contributes $m(m-1)(m-2)$ once; a pair of mult m contributes twice that — its two zeros). Consequences:

1. *On the class $\{$mark-$\{1,2\}$ atoms$\} \cup \{$multiplicity-1 pairs of arbitrary depth$\}$ — the entire doubles-vs-pairs comparison class of `lemmaR_tight` — the right side vanishes identically: $C = 3F - 2\,\mathrm{Mass}$ identically on the class. The signed cubic row restricted to isolated blocks of this class is an exact affine combination of the Frobenius and mass rows: it separates nothing those two rows do not. An on-line double and an off-line pair of any depth are cubic-indistinguishable.*

2. *All discriminating power of the cubic row beyond the two-moment data therefore lives in exactly two places: (i) marks ≥ 3 (and pair multiplicities ≥ 2), through the nonnegative integrality terms on the right; (ii) violations of block isolation — the clustering cross-terms of the master trace formula, which the identity does not constrain, and which Theorem 3.9 shows to be adversary-tunable at zero cost.*

3. *The premise "a hyperbolic pair block contributes $(+a)^3 + (-a)^3 = 0$ to $\operatorname{tr} G^3$ while a double contributes $+8$" is false under the true block law: the pair's signed cubic is $2m^3(1+3A^2) \geq 8m^3$ at every depth. The sign-split "different corners of the constraint polytope" picture is superseded.*

Proof. Sum the per-block identities. An atom of mark m has block charges (mass m, Frobenius m^2, cubic m^3), contributing $m^3 - 3m^2 + 2m = m(m-1)(m-2)$ to $C - 3F + 2\,\mathrm{Mass}$. A pair of mult m has block charges (mass $2m$, Frobenius $2m^2(1 + A^2)$, cubic $2m^3(1 + 3A^2)$), contributing

$$2m^3(1 + 3A^2) - 6m^2(1 + A^2) + 4m = 2m(m-1)(m-2) + 6m^2(m-1)A^2\,,$$

exactly its two zeros' worth of the integrality sum plus the multiplicity-weighted depth term. On marks $\{1, 2\}$ and mult-1 pairs every term vanishes. Consequence 3 is Proposition 3.2 ($A \geq 1$). $\square$

In classical language this is the Chebyshev–Markov–Stieltjes phenomenon that an odd moment does not improve a one-sided bound in a Chebyshev system (Karlin and Studden [KS66]); Theorem 3.4 is its exact form for the block class at issue here.

Remark (why the row system is clustered). By consequence 1, a linear program built on isolated blocks over the doubles-vs-pairs class returns a tautological answer along the plane $c = 3F - 2$. The clustered null budgets and positional freedom of Section 2.5 are forced rather than conservative, and the clustering cross-terms, the only live channel, are a free resource for the adversary rather than a signal for the certificate.

Numerical confirmation (`verify_t3.py`): charges and the per-zero identity for $m \in \{1, 2, 3, 5\}$ at all depths to $\leq 9.1 \times 10^{-13}$; atom identity exhaustive $m = 1, \ldots, 29$, exact; master-formula 2×2 kernel route reproduces all three charges.

3.3 Sine-process closed forms and the doubles-plane diagnostic

The null budgets are the per-zero moments of the flat-window Gram over the determinantal sine process — the process with kernel $S(x) = \sin(\pi x)/(\pi x)$, intensity 1, and correlation functions (taken as the null's definition)

$$\begin{aligned}
\rho_2(x_1, x_2) &= 1 - S(x_{12})^2,\\
\rho_3(x_1, x_2, x_3) &= 1 - S(x_{12})^2 - S(x_{13})^2 - S(x_{23})^2 + 2S(x_{12})S(x_{13})S(x_{23}).
\end{aligned}$$

The per-zero Gram moments at bandwidth λ (flat window, $\psi = \psi_\lambda$, $u = \widehat{\psi}$):

$$\begin{aligned}
m_2(\lambda) &:= 1 + \int \rho_2(0, x)\, \psi(x)^2\, dx\,,\\
m_3(\lambda) &:= 1 + 3\int \rho_2(0, x)\, \psi(x)^2\, dx + T_3(\lambda)\,,\\
T_3(\lambda) &:= \iint \rho_3(0, x, x + y)\, \psi(x)\psi(y)\psi(x + y)\, dx\, dy\,,
\end{aligned}$$

the $N \to \infty$ limits of $E[\operatorname{tr} G^k]/N$: the k-tuple sum of Section 2.2 splits over coincidence patterns — all-equal (value 1), exactly-two-equal (3 ordered patterns at $k = 3$, each $\psi(x)^2$ against ρ_2), all-distinct (ρ_3 against the cyclic kernel product, reparametrized by separations using the evenness of ψ). The sine process is simple, so all marks are 1.

Theorem 3.5 (flat-window closed forms)**.** *For the flat window and every* $0 < \lambda \leq 1$*:*

$$m_2(\lambda) = \frac{1}{\lambda} + \frac{\lambda}{3}\,, \qquad m_3(\lambda) = 1 + \frac{1}{\lambda^2}\,.$$

Proof. Step 1 (Fourier-side reduction). With $t(\alpha) = (1 - |\alpha|)_+$ (the transform of S^2), $b = \mathbf{1}_{[-1/2,1/2]}$ $(= \widehat{S})$, and the convention $\widehat{f}(\alpha) = \int f(x) e^{-2\pi i \alpha x}\, dx$:

(i) $\int \psi^2\,dx = \int u^2\,d\alpha$ [Plancherel];

(ii) $\int S^2\psi^2\,dx = \int t\,(u*u)\,d\alpha$ [Plancherel on the product pair];

(iii) $\iint \psi(x)\psi(y)\psi(x+y)\,dx\,dy = \int u^3\,d\alpha$;

(iv) $\iint S(x)^2\psi(x)\psi(y)\psi(x+y)\,dx\,dy = \int (t*u)\,u^2\,d\alpha$, and the same value for the $S(y)^2$ and $S(x+y)^2$ variants;

(v) $\iint S(x)S(y)S(x+y)\psi(x)\psi(y)\psi(x+y)\,dx\,dy = \int (b*u)^3\,d\alpha$.

For (iii), two Plancherel steps with no delta calculus: with y fixed, $g(y) := \int \psi(x)\psi(x+y)\,dx$ is the Fourier transform of u^2 at y; then $\int \psi(y)g(y)\,dy = \int \widehat{\psi}\,u^2 = \int u^3$ by the multiplication identity $\int f\widehat{g} = \int \widehat{f}g$; Fubini is licensed by $|\psi(x)\psi(y)\psi(x+y)| \le C/((1+|x|)(1+|y|)(1+|x+y|))$, integrable on $\mathbf{R}^2$. For (iv), the same two-step Plancherel with first factor $S(x)^2\psi(x)$ gives $\int (S^2\psi)\widehat{\ }\,u^2 = \int (t*u)u^2$; the $S(y)^2$ and $S(x+y)^2$ variants reduce to the same value under the substitutions $(x,y)\mapsto(y,x)$ and $(x,y)\mapsto(x,-x-y)$ (Jacobian 1, integrand invariant). For (v), set $h = S\psi$, $\widehat{h} = b*u$, and apply (iii) to h. Hence

$$m_2 = 1 + \int u^2 - \int t\,(u*u), \qquad m_3 = 1 + 3\Big[\int u^2 - \int t\,(u*u)\Big] + T_3,$$
$$T_3 = \int u^3 - 3\int (t*u)u^2 + 2\int (b*u)^3.$$

Step 2 (evaluation at $u = (1/\lambda)\mathbf{1}_{[-\lambda/2,\lambda/2]}$, using $\lambda \le 1$ exactly where indicated).

(1) $\int u^2 = 1/\lambda$.

(2) $(u*u)(\alpha) = (1/\lambda^2)(\lambda - |\alpha|)_+$; since $\lambda \le 1$, $t = 1 - |\alpha|$ on this support, so $\int t\,(u*u) = (2/\lambda^2)\int_0^\lambda (1-a)(\lambda-a)\,da = (2/\lambda^2)[\lambda^2/2 - \lambda^3/6] = 1 - \lambda/3$. Hence $m_2 = 1 + 1/\lambda - (1-\lambda/3) = 1/\lambda + \lambda/3$.

(3) $\int u^3 = 1/\lambda^2$.

(4) $\int (t*u)u^2 = (1/\lambda^3)\iint_{|\alpha|\le\lambda/2,\ |\alpha-\beta|\le\lambda/2} t(\beta)\,d\beta\,d\alpha = (1/\lambda^3)\int t(\beta)(\lambda-|\beta|)_+\,d\beta = (1/\lambda^3)[\lambda^2 - \lambda^3/3] = 1/\lambda - 1/3$ (reusing the integral of (2); valid for $\lambda \le 1$).

(5) $(b*u)(\alpha)$ is the trapezoid equal to 1 on $|\alpha| \le (1-\lambda)/2$, decaying linearly to 0 at $|\alpha| = (1+\lambda)/2$; hence $\int (b*u)^3 = (1-\lambda) + 2\lambda\int_0^1 s^3\,ds = 1 - \lambda/2$.

Assembling: $T_3 = 1/\lambda^2 - 3(1/\lambda - 1/3) + 2(1-\lambda/2) = 1/\lambda^2 - 3/\lambda + 3 - \lambda$, and $m_3 = 1 + 3[1/\lambda - 1 + \lambda/3] + T_3 = 1 + 1/\lambda^2$. □

Corollary 3.6 (anchor values, the doubles-plane diagnostic, the moment margin)**.**

$$m_2(1) = 4/3, \quad m_3(1) = 2, \quad m_2(1/2) = 13/6, \quad m_3(1/2) = 5.$$

Define $G(\lambda) := m_3 - (3m_2 - 2)$ (the null budget's excess over the isolated-block doubles/pairs plane of Theorem 3.4) and $\mathrm{Marg}(\lambda) := 2m_2 - m_3$ (the certificate margin of [AF26, §7.3]). Then, flat window, on $(0,1]$:

$$G(\lambda) = 3 + \frac{1}{\lambda^2} - \frac{3}{\lambda} - \lambda\,; \qquad G(1) = 0, \qquad G(1/2) = 1/2,$$
$$\mathrm{Marg}(\lambda) = \frac{2}{\lambda} + \frac{2\lambda}{3} - 1 - \frac{1}{\lambda^2}\,; \qquad \mathrm{Marg}(1) = 2/3, \qquad \mathrm{Marg}(1/2) = -2/3,$$

and Marg *has a unique sign change on* $(0,1]$ *(the interval on which the closed forms are proved), at* $\lambda_0 = 0.610511\ldots$ *Each value is load-bearing for the no-go:*

1. $G(1) = 0$ ***is the "$2 = 2$" saturation.*** *At bandwidth one the sine budget* $m_3(1) = 2$ *exactly equals the isolated-block cubic demand of the* $5/6$*-extremal (*$2/3$ *simple +* $1/6$ *doubles:* $(2/3)(1) + (1/3)(4)$ *mass-weighted* $= 2$*, on the plane of Theorem 3.4). The bandwidth-one cubic row alone cannot cut the doubles corner — the budget is exactly met.*

2. $G(1/2) = +1/2 > 0$*. At the operating bandwidth the null budget sits strictly above the doubles plane: a doubles-saturated adversary must supply* $+1/2$ *per zero of cubic through positional cross-terms — which Theorem 3.9's grid freedom supplies at zero cost (the witness of Section 4.4 does exactly this, sitting at the lower cubic budget edge).*

3. $\mathrm{Marg}(1/2) = -2/3 < 0$*. The* $\omega(m)$*-moment route's margin is negative at the proven operating point: a strictly positive marginal value could never have come from the mechanism of [AF26, §7.3] there, only from joint positional pricing. The sign change at* $\lambda_0 = 0.6105$ *is precisely where the exploratory scan saw* δ_0 *turn positive (between* $\lambda' = 0.55$ *and* 0.60*; Section 8.4), outside the proven ladder regime.*

Proof. Substitute Theorem 3.5; the displayed formulas are algebraic identities on $(0,1]$. $G(1) = 3+1-3-1 = 0$; $G(1/2) = 3+4-6-1/2 = 1/2$; $\mathrm{Marg}(1) = 2+2/3-1-1 = 2/3$; $\mathrm{Marg}(1/2) = 4+1/3-1-4 = -2/3$. Uniqueness of the root: $q(\lambda) = \lambda^2\,\mathrm{Marg}(\lambda) = 2\lambda + (2/3)\lambda^3 - \lambda^2 - 1$ has $q' = 2 + 2\lambda^2 - 2\lambda > 0$ (negative discriminant), so q is strictly increasing with $q(0.55) < 0 < q(0.66)$; numerically the root is 0.610511 (`brentq`, re-verified by `sympy nsolve`). The extremal demand in item 1: per-zero $(F, c) = (1, 1)$ on simples and $(2, 4)$ on doubles; mass-weighted cubic $(2/3)(1) + (1/3)(4) = 2 = m_3(1)$ and Frobenius $(2/3)(1) + (1/3)(2) = 4/3 = m_2(1)$ — both budgets exactly saturated at $\lambda = 1$. □

Corollary 3.7 (the $m_2 = \kappa$ identity: sine null = unconditional budget)**.** *The closed form* $m_2(\lambda) = 1/\lambda + \lambda/3$ *coincides, at every* $\lambda \le 1$*, with the value obtained by integrating the unconditional Montgomery pair-correlation form factor of [BGSTB24],* $F(\alpha) = \delta_{\mathrm{D}}(\alpha) + |\alpha|$ *(on* $|\alpha| \le 1$*),* δ_{D} *being the Dirac mass at the origin — not the marginal value* δ_0 *— against the flat-window pair kernel:*

$$\frac{1}{\lambda^2}\int_{-\lambda}^{\lambda}(\lambda - |\alpha|)\big[\delta_{\mathrm{D}}(\alpha) + |\alpha|\big]\,d\alpha = \frac{\lambda + \lambda^3/3}{\lambda^2} = \frac{1}{\lambda} + \frac{\lambda}{3} = m_2(\lambda)\,.$$

The sine-process null's second-moment budget is the unconditional prime-side $\kappa(\lambda)$ *of Alpöge and Furman, exactly, on the whole band* $\lambda \le 1$ *— an exact identity check at second order; the* F' *budget row is therefore backed by unconditional data. (Provenance: the prime-side member —* $\mathrm{tr}\,\widehat{G}^2 \leftrightarrow \int(\lambda - |\alpha|)F(\alpha)\,d\alpha$ *with* F *unconditional and pointwise on the closed band* $0 \le \alpha \le 1$ *— is the Theorem 5.8 / Remark 5.10 bookkeeping of [AF26] together with Theorem 1 of [BGSTB24], quoted, not re-derived; the displayed arithmetic identity is the elementary integral* $\int_0^\lambda(\lambda - a)a\,da = \lambda^3/6$*, doubled, plus the delta term* λ*.) No third-order analog is claimed: the identification* $c_3(\lambda') = m_3(\lambda')$ *rests on Rudnick–Sarnak-range GUE* 3*-level correlations and remains open (the cubic budget center; Section 8.3).*

Numerical confirmation (`verify_t4.py`, `verify_t4_mc.py`): Fourier-side quadrature matches both closed forms at 7 bandwidths to $\le 1.7 \times 10^{-5}$; an independent real-space determinantal quadrature gives $m_3(1/2) = 4.9924$ (truncation $X = 40$) $\to 4.9960$ ($X = 80$), trending to 5 as a second independent quadrature did ($4.9896 \to 4.9932$); a third independent sampler, CUE Monte

Carlo, extrapolates $m_2(1/2) \to 2.1668$ and $m_3(1/2) \to 5.0014$ against 13/6 and 5, each within $\sim 1\sigma$ of the other two (full tables: Appendix B). The values $m_2(1) = 4/3$, $m_3(1) = 2$ also discharge, for $k \le 3$, the sine moments $m_k(1) = 1, 4/3, 2$ recalled in [AF26].

3.4 Grid Parseval decoupling

The absorption mechanism rests on one exact identity: on a specific uniform grid the entire bandwidth-one Frobenius row degenerates to the multiplicity count — for every site subset and every mark assignment — while the half-band kernel on the same grid remains position-sensitive.

Lemma 3.8 (flat-band lemma)**.** *Let $M \ge 1$ and let the harmonic band be any set of M consecutive integers $B = \{j_0, \dots, j_0 + M - 1\}$, with uniform weights $u_j = 1/M$. Let the configuration be supported on the M-site uniform grid $\theta_k = kN/M$ ($k = 0, \dots, M-1$) on the circle of circumference N, with arbitrary real marks $m_k \ge 0$. Then*

$$\operatorname{tr} \widehat{G}^2 = \sum_k m_k^2 \qquad \textit{exactly.}$$

Proof. On the grid the form factor is a DFT on $\mathbf{Z}_M$: $c_s = \sum_k m_k e^{-2\pi i s k/M}$, which depends only on $s \bmod M$ — c is M-periodic in s — and DFT Parseval gives $\sum_{r \bmod M} |c_r|^2 = M \sum_k m_k^2$ (P). The Frobenius row is $\sum_s W_2(s)|c_s|^2$ with $W_2(s) = \#\{(j_1, j_2) \in B \times B : j_1 + j_2 = s\}/M^2$; the sum-set $B + B = \{2j_0, \dots, 2j_0 + 2M - 2\}$ carries the triangular pair counts $M - |s - (2j_0 + M - 1)|$ on $|s - (2j_0 + M - 1)| \le M - 1$. Fix a residue class $r \bmod M$: the window $B + B$ has length $2M - 1$, so it contains either one member of the class ($t = 0$, count M) or two members t and $t \mp M$ with $0 < |t| < M$, whose counts sum to $(M - |t|) + |t| = M$ — exactly M on every residue class. Hence, by the M-periodicity of $|c_s|^2$,

$$\operatorname{tr} \widehat{G}^2 = \frac{1}{M^2} \sum_{r \bmod M} M |c_r|^2 = \frac{1}{M} \sum_r |c_r|^2 = \sum_k m_k^2$$

by (P). $\square$

Theorem 3.9 (grid Parseval decoupling at bandwidth one)**.** *Let N be even, flat window at $\lambda = 1$: band $B = \{|j| \le N/2\}$, $M = N + 1$ harmonics, $u_j = 1/(N+1)$. Let the configuration be supported on the $(N+1)$-site uniform grid of spacing $N/(N+1)$ (the ψ_1-zero grid), with arbitrary marks. Then*

$$F_1 = \operatorname{tr} \widehat{G}_1^2 = \sum_z m_z^2 \qquad \textit{exactly,}$$

for every site subset and every mark assignment: the bandwidth-one reading on this grid sees only multiplicities, never positions, and the `lemmaR_tight` extremal analysis transfers verbatim to grid configurations.

Proof. Lemma 3.8 with $M = N + 1$, $B = \{-N/2, \dots, N/2\}$ ($N + 1$ consecutive integers since N is even), grid spacing $N/M = N/(N+1)$. $\square$

Nomenclature. Lemma 3.8 is critical-sampling Parseval for a flat band — the discrete orthogonality of M consecutive characters on an M-site uniform grid — and Proposition 3.11 is its Shannon–Whittaker continuum form on the integer lattice. Both are classical; they are proved here in the exact finite-circle normalization of Section 2.3, and the Lean development machine-checks

Lemma 3.8 over an arbitrary integral domain (Section 10). The degeneracy they create at critical sampling is the one Lagarias and Rodgers study in [LR21]: above the Nyquist bandwidth the mimicking correlations are unique, at critical sampling they are not. The grid used here (spacing $a = N/(N+1)$, band $B = 1/2$) falls in the region their Theorem 1.8 leaves undetermined, and the witness of Section 4.4 matches finitely many trace rows rather than all correlations, so there is no conflict; their negative results are why the absorption cannot be upgraded to full n-level mimicry.

Remark (why this grid). Equivalently: all pairwise site differences are nonzero multiples of $N/(N+1)$, the zero set of the discrete bandwidth-one kernel D. The Fourier proof makes the exactness structural — assembly weights constant on residue classes — rather than a numerical coincidence.

Proposition 3.10 (the $\lambda' = 1/2$ kernel is alive on the same grid)**.** *Same geometry, N and $N/2$ even, $\lambda' = 1/2$: band $B' = \{|j| \le N/4\}$, $M' = N/2 + 1$. For grid-supported configurations,*

$$F' = \sum_{|s| \le N/2} \frac{M' - |s|}{M'^2} |c_s|^2 ,$$

where $\{-N/2, \ldots, N/2\}$ is a complete residue system mod $(N+1)$ and c_s is the $\mathbf{Z}_{N+1}$ DFT of the mark vector. The weights are strictly positive and non-constant across residues (triangular: at $N = 64$, from $1/33$ at $s = 0$ down to $1/33^2$ at $|s| = 32$), so F' is a strictly position-sensitive functional of the grid configuration, subject only to Parseval.

Proof. $F' = \sum_s W_2'(s)|c_s|^2$ with $W_2'(s) = (M' - |s|)_+/M'^2$ supported in $|s| \le N/2$. That range has $N+1$ elements, hence meets each residue class mod $N+1$ exactly once; no residue-class telescoping occurs (band length $M' = N/2+1$ differs from the grid size $N+1$), so F' reads the full grid power spectrum with a non-uniform weight; distinct site subsets of equal mark statistics generically differ. □

Worked exact value: the witness's first column. $N = 64$, the "vacancy lattice": 64 simple marks on 64 of the 65 grid sites. Then $c_0 = 64$ and $|c_s|^2 = 1$ for $s \not\equiv 0 \bmod 65$, so

$$\begin{aligned} F_1 &= 64 \qquad \text{(Theorem 3.9)}, \\ F' &= \frac{33}{33^2}\, 64^2 + \sum_{0<|s|\le 32} \frac{33 - |s|}{33^2} = \frac{4096}{33} + \frac{32}{33} = \frac{4128}{33} = 125.0909\ldots , \end{aligned}$$

using $\sum_{|s|\le 32}(33 - |s|) = 33^2$ and subtracting the $s = 0$ term. This reproduces the certified witness column exactly (`witness_N64.json`: $F_1 = 64$, $F' = 125.09090\ldots$). The two-bandwidth decoupling is thereby explicit: F_1 is blind to the vacancy's position, F' is not.

Proposition 3.11 (continuum analog: the adversary lattice)**.** *In the continuum model, flat window, configuration supported on the integer lattice $\mathbf{Z}$ with arbitrary real marks: $\operatorname{tr} G_1^2 = \sum m_z^2$ exactly, while the half-band kernel is alive on the lattice: $\psi_{1/2}(k) = \operatorname{sinc}(\pi k/2) = (2/(\pi k))(-1)^{(k-1)/2}$ for odd k (0 for even $k \ne 0$). For the finite-T Weil–Gabor matrix itself the identity holds as $\operatorname{tr} \widehat{G}^2 = (\sum m^2)(1 + o(1))$, the $o(1)$ being the finite-window corrections of [AF26] (quoted).*

Proof (model part). At $k = 2$ the master formula gives $\operatorname{tr} G_1^2 = \sum_{z,z'} m_z m_{z'} \psi_1(z - z')^2$; $\psi_1(x) = \operatorname{sinc}(\pi x)$ vanishes at every nonzero integer and equals 1 at 0, so only the diagonal survives. The half-band values are $\sin(\pi k/2)/(\pi k/2)$ with $\sin(\pi k/2) = (-1)^{(k-1)/2}$ for odd k. □

Remark. Theorem 3.9 is the exact periodization of Proposition 3.11: spacing $N/(N+1)$ instead of 1 makes the identity exact rather than asymptotic, which is why the finite-N computation exhibits $\delta_0 = 0$ as an exact LP equality rather than a numerical near-zero. The absorption is not a finite-N artifact.

Numerical confirmation (`verify_t1.py`): grid exactness on 40 random site-subsets with random marks at $N = 64$ and $N = 128$, two independent evaluation routes (Fourier double sum; eigenvalues of H): $\max|F_1 - \sum m^2| = 9.1\times10^{-13}/4.1\times10^{-12}$, matching the independently computed 4.5×10^{-13}; general flat-band lemma on 20 random (N, M, j_0) draws including asymmetric bands: 3.2×10^{-12}; controls — off-grid configurations give $|F_1 - \sum m^2| \in [1.3 \times 10^{-2},\ 62.5]$ and on-grid half-band $|F' - \sum m^2| \in [40.9, 120.8]$ (the identity genuinely fails off the grid, and the half-band kernel is genuinely alive on it); continuum lattice exact to machine precision.

4 The absorption theorem

This section states and proves the model-level theorem the LP pins — the two-moment data force exactly the `lemmaR_tight` corner, for both benchmarks, over the model's configuration classes — and then gives the computational certification that the entire cubic block is absorbed at that corner, with the explicit witness.

4.1 The atom-only corner theorem

Fix N and $\varepsilon \geq 0$, and let $\mathcal{A}$ be the class of atom-only configurations: atoms at arbitrary positions (finite-circle or continuum model, flat windows — the proofs work verbatim in both), integer marks $m_i \geq 1$, mass $\sum m_i = N$ per period.

Lemma 4.1 (diagonal positivity)**.** *For every atom-only configuration (any positions, any real marks $m_i \geq 0$), in either model, $F_1 \geq \sum_i m_i^2$, with equality iff every off-diagonal kernel value vanishes — in particular on the grid of Theorem 3.9 and the lattice of Proposition 3.11.*

Proof. $F_1 = \sum_{i,i'} m_i m_{i'} K(\theta_i - \theta_{i'})$ with $K = D^2$ (finite circle) or $K = \psi_1^2$ (continuum): $K \geq 0$ pointwise, $K(0) = 1$, so $F_1 = \sum m_i^2 +$ (a sum of nonnegative off-diagonal terms). □

This is the one step that fails for pair-containing configurations — an off-line pair contributes cross-terms of the form $2mm' \operatorname{Re}[\psi(x+iy)^2]$, which need not be nonnegative; Sections 4.2–4.3 treat that channel.

Lemma 4.2 (integer-mark corner inequalities)**.** *For every atom-only configuration with integer marks $m_i \geq 1$ and mass N:*

$$\begin{aligned} &(a) \quad N_d = \#\text{atoms} \;\geq\; \frac{3N - \sum_i m_i^2}{2}\,, \\ &(b) \quad n_1 = \#\textit{mark-}1 \text{ atoms} \;\geq\; 2N - \sum_i m_i^2\,, \end{aligned}$$

each with equality iff all marks lie in $\{1, 2\}$.

Proof. Per-atom inequalities summed over atoms. (a) For integer $m \geq 1$, $(m-1)(m-2) \geq 0$, i.e. $1 \geq (3m - m^2)/2$, equality iff $m \in \{1, 2\}$; sum. (b) $\mathbf{1}_{m=1} \geq 2m - m^2$: equality at $m = 1, 2$; at $m \geq 3$ the right side is negative; sum. □

Remark. (a) is the second integrality level $(m-1)(m-2) \geq 0$ — exactly the level `lemmaR_tight` proves exhausted by the two-moment certificate, and, by Theorem 3.4, exactly the level the cubic row re-measures on isolated blocks; (b) is its simple-fraction sibling. The device is Conrey, Ghosh and Gonek's [CGG98], made unconditional in §7.2(c) of [AF26]. In the terms of that paper, (a) is the doubles-optimal corner arithmetic: an atom of mark m spends $m(m-1)$ of Frobenius excess to save $m-1$ units of N_d, so doubles are the efficient coin.

Theorem 4.3 (atom-only corner theorem, both benchmarks)**.** *Fix N even, $0 \leq \varepsilon \leq 1/2$ (finite-circle model, flat windows). Over laws w on $\mathcal{A}$ with mass N and $E_w[F_1] \leq (4/3)N(1+\varepsilon)$:*

$$\begin{aligned} &\textit{(a)} \quad \min_w \; E_w[N_d]/N = \frac{5}{6} - \frac{2}{3}\varepsilon\,, \\ &\textit{(b)} \quad \min_w \; E_w[p_1] = 2 - \frac{4}{3}(1+\varepsilon) = \frac{2}{3} - \frac{4}{3}\varepsilon\,. \end{aligned}$$

Both minima are attained by explicit laws supported on the $(N+1)$-site grid of Theorem 3.9 with marks $\{1,2\}$ only. Every configuration in $\mathcal{A}$ satisfies the pointwise versions with its own Frobenius excess ε_c. The same statements hold with the two-sided budget band $(4/3)N[1-\varepsilon, 1+\varepsilon]$ (the attaining laws sit at the upper edge), and the minima over any richer mark alphabet (any W, or unbounded marks) are the same.

Proof. Lower bounds: by Lemma 4.1, $E_w[\sum m^2] \leq E_w[F_1] \leq (4/3)N(1+\varepsilon)$; by Lemma 4.2(a) and linearity, $E_w[N_d] \geq (3N - (4/3)N(1+\varepsilon))/2 = N(5/6 - (2/3)\varepsilon)$; by Lemma 4.2(b), $E_w[n_1] \geq 2N - (4/3)N(1+\varepsilon) = N(2/3 - (4/3)\varepsilon)$.

Attainment: for integer $0 \leq k \leq N/2$ let c_k be the grid configuration with k doubles and $N-2k$ simples on $N-k$ distinct grid sites (which exist: $N+1 \geq N-k$). By Theorem 3.9, $F_1(c_k) = N+2k$ exactly; $N_d(c_k) = N-k$; $n_1(c_k) = N-2k$. Set $\bar{k} = (B-N)/2$ with $B = (4/3)N(1+\varepsilon)$; then $0 \leq \bar{k} \leq N/2$ for $\varepsilon \leq 1/2$. The two-column law w^* mixing $c_{\lfloor \bar{k} \rfloor}$ and $c_{\lfloor \bar{k} \rfloor + 1}$ with $E[k] = \bar{k}$ achieves

$$\begin{gathered} E[F_1] = N + 2\bar{k} = B \quad \text{(upper edge, exactly)}, \\ \frac{E[N_d]}{N} = \frac{3N-B}{2N} = \frac{5}{6} - \frac{2}{3}\varepsilon, \qquad E[p_1] = 2 - \frac{B}{N} = \frac{2}{3} - \frac{4}{3}\varepsilon\,, \end{gathered}$$

and both lower-bound chains hold with equality (marks in $\{1,2\}$; grid kills all cross-terms; budget saturated), so w^* is simultaneously optimal for both objectives. The two-sided band adds only the lower budget edge, which w^* satisfies. Enlarging the mark alphabet cannot lower the minimum (the chain never used a mark cap) and cannot raise it (w^* uses marks $\{1,2\}$, inside every alphabet): the W-independence is inclusion, not extrapolation. □

Corollary 4.4 (asymptotic and matched corners)**.** *As $\varepsilon \to 0$ the two optima tend to $5/6$ and $2/3$ — the model's bandwidth-one analogs of Montgomery's extremal values [Mon73] for distinct zeros and simple fraction. With the matched-geometry budget B_1 replacing $(4/3)N$, the same proof gives $\min E[N_d]/N = (3N - B_1(1+\varepsilon))/(2N)$ and $\min E[p_1] = 2 - (B_1/N)(1+\varepsilon)$; at the matched values used here ($N = 64$, $\varepsilon = 0.05$, $B_1 = 84.71214548308222$) these are 0.8050957 and 0.6101914 — the computed optima to LP precision.*

Remark (a numerical coincidence, disambiguated). The corner value 0.8050957 rounds to 0.8051 at four decimals, and the string 0.8051 also occurs in the zeta literature as Farmer–Gonek–Lee's RH-conditional pair-correlation lower bound for the proportion of *distinct* zeros of zeta, $N_d(T) > 0.8051\,N(T)$ [FGL14, Cor. 1.4] (since superseded by 0.84665 in [BHB13] and by 0.8477 on RH / 0.8486 on GRH in [CGdL20, Cor. 3]). The two are unrelated: the value here is a finite-N LP corner of the bandwidth-one model at $N = 64$, $\varepsilon = 0.05$, $B_1 = 84.71214548308222$, and carries no information about zeta.

Numerical confirmation (`verify_t2.py`): per-atom inequalities exhaustive to $m = 50$ plus 20,000 random draws (0 violations); grid-column LP matches the closed forms at $\varepsilon \in \{0.10, 0.05, 0.02, 0.002\}$ to $\leq 3.3 \times 10^{-16}$ for both objectives; exact rational identities verified in `Fraction` arithmetic; the certified witness reproduces $P = (3N - E[F_1])/(2N) = 0.8050957$ to 6×10^{-13}.

4.2 Integrality is necessary: the exact fractional counterexample

Any extension of Theorem 4.3 beyond the atom-only class must consume integrality, as the following counterexample shows. Work in the primary geometry ($N = 64$, flat window, $\lambda = 1$; kernel $\psi(z) = \sum_j u_j e^{-2\pi i j z/N}$, grid $g_k = k\,64/65$) and define $\bar{a}(x) := \psi(ix) = \sum_j u_j \cosh(2\pi j x/N) \geq 1$. Write $M(c)$ for total mark mass, $S_2(c) = \sum_z m_z^2$ (each pair member counted m^2), $T(c) = 3M(c) - 2N_d(c)$, and consider the *master inequality*

$$\text{(MI)} \qquad F_1(c) \;\geq\; 3M(c) - 2N_d(c)\,,$$

which is precisely what the corner consumes: (MI) per configuration gives, for every law with mass N and $E[F_1] \leq (4/3)N(1+\varepsilon)$, both corners — $E[N_d]/N \geq 5/6 - (2/3)\varepsilon$ and $E[p_1] \geq 2/3 - (4/3)\varepsilon$. (Per configuration (MI) reads $N_d \geq (3M - F_1)/2$; take expectations. For simples, $\mathbf{1}_{m=1} \geq 2 - m$ summed over the zero multiset gives $n_1 \geq 2N_d - M$, hence $E[n_1] \geq (3N - E[F_1]) - N$.) Note $S_2 \geq T$ for integer marks (per zero, $m^2 - (3m - 2) = (m-1)(m-2) \geq 0$), so (MI) is weaker than $F_1 \geq S_2$; the closure arguments below target (MI) because it carries the integrality slack explicitly.

Proposition 4.5 (fractional marks break the floor at every depth)**.** *Let c_μ be the vacancy lattice (unit atoms on grid sites $g_1, \ldots, g_{64}$) plus one pair at the hole g_0 with depth $d > 0$ and real mark $\mu > 0$. Then, exactly,*

$$F_1(c_\mu) - S_2(c_\mu) = 2\mu^2 \bar{a}(2d)^2 - 4\mu\,(\bar{a}(d)^2 - 1),$$

which is negative for every $0 < \mu < 2(\bar{a}(d)^2 - 1)/\bar{a}(2d)^2$ — the pair harvests the hole's missing coupling linearly in μ while paying only quadratically. For integer marks the same family is safe: at $\mu = m \in \mathbf{Z}$, the Cauchy–Schwarz bound $\bar{a}(d)^2 - 1 \leq (\bar{a}(2d) - 1)/2$ gives $F_1 - S_2 \geq 2m^2\bar{a}(2d)^2 - 2m(\bar{a}(2d) - 1) > 0$.

Proof. On the vacancy lattice the atom form factor is $\varphi_0(s) = 64$ at $s \equiv 0$ and -1 otherwise; the pair adds $2\mu\cosh(\beta s)$, $\beta = 2\pi d/N$. Expanding $F_1 = \sum_s w_s |c_s|^2$ with the generating identities $\sum_s w_s \cosh(\beta s) = \bar{a}(d)^2$ and $\sum_s w_s \cosh(2\beta s) = \bar{a}(2d)^2$ (the assembly weights $w = u * u$ factor through ψ at imaginary arguments) gives the display; the numeric check at $(d, \mu) = (0.25, 0.05)$ reproduces it to 10^{-12}, with $F_1 - S_2 = -3.520 \times 10^{-2}$. The integer case uses Cauchy–Schwarz on the positive weights u_j, which sum to 1: $\bar{a}(d)^2 = \big(\sum_j u_j \cosh(2\pi j d/N)\big)^2 \leq \sum_j u_j \cosh^2(2\pi j d/N) = (1 + \bar{a}(2d))/2$, the last equality by $\cosh^2 t = (1 + \cosh 2t)/2$. $\square$

Reading. An adversary with fractional marks takes $\mu \to 0$ and wins at any depth; integer marks force $\mu \geq 1$, where the quadratic cost dominates. Any closure of the pair channel must therefore consume integrality — the channel is invisible to every convexified or soft-positivity analysis. (This is the textbook *integrality gap* in its usual form: the fractional relaxation of the marked-configuration program has a strictly smaller optimum than the integer program, so a bound proved for the relaxation is not a bound for the problem.) It also explains the LP's phenomenology:

its columns are explicit integer configurations (nothing about marks is relaxed), which is why its converged pricing found no pair column. It is likewise the sharp form of the realizability discussion of Section 2.4: with fractional marks even the 5/6 baseline is false.

4.3 The pair-interference channel: closed on the pair-depth family (multi-pair to $w \le 0.98$), backstopped everywhere

The full closure argument, with its capacity machinery and certified constants, is given in the companion note `results/a4-no-go/pair-channel.md` of the repository (see *Provenance, data and code* at the end of this paper). This subsection states the results, the exact ledger they rest on, and the proofs that are short enough to give in full. Everything is at the primary geometry ($N = 64$, flat window, $\lambda = 1$); depths d are in circle units, $w = 2\pi d$ dimensionless (the admissible pair family of Section 2.4 is $w \in (0, 1]$ plus deep probes $w \in \{1.5, 2\}$, i.e. $d \le 0.3183$).

The exact interference ledger. For every admissible configuration (any atoms, any pairs, any integer marks), with $R_y(x) := \operatorname{Re} \psi(x + iy)^2$:

$$\begin{aligned}
F_1 - T = {} & \sum_a (m_a - 1)(m_a - 2) + 2 \sum_p (m_p - 1)(m_p - 2) && \text{[integrality slack]} \\
& + \sum_{a \ne a'} m_a m_{a'} \psi(x_{aa'})^2 && \text{[atom-atom} \ge 0\text{]} \\
& + \sum_p 2 m_p^2 \bar{a}(2d_p)^2 && \text{[pair diagonals]} \\
& + \sum_p 4 m_p \sum_a m_a R_{d_p}(x_{pa}) && \text{[pair-atom]} \\
& + \sum_{p<q} 4 m_p m_q \big[R_{d_p + d_q}(x_{pq}) + R_{|d_p - d_q|}(x_{pq}) \big] && \text{[pair-pair]}
\end{aligned}$$

(verified on 25 random mixed configurations to 2.2×10^{-11}, the stored `pairchan_verify_out.json` maximum). Everything on the right is nonnegative except the R-couplings, whose negative parts live only in the dips of R_y; the question is whether the dips can outrun the pair diagonals plus the integrality slack. They cannot: at the worst depth the margin is a factor of about 1.5.

Theorem 4.6 (single-pair closure)**.** *Every configuration with exactly one pair (arbitrary atoms, arbitrary integer marks, arbitrary positions) satisfies (MI) whenever the pair depth is $d \le 0.45$ — i.e. $w \le 2.8$, covering the entire admissible pair-depth family including both deep probes, with $\ge 34\%$ capacity margin on the family itself (certified stacking-corrected capacity ratio ≤ 0.66 on the family, ≤ 0.921 out to $d = 0.45$). For mass $\le N$ configurations (the LP class) (MI) also holds for every $d \ge 1.0$; the window $(0.45, 1.0)$ is certified numerically (raw capacity ratio ≤ 0.6823 throughout; direct adversarial searches reach at most 74% of the pair budget).*

Theorem 4.7 (multi-pair closure)**.** *(MI) holds for every configuration all of whose pair depths satisfy $w \le 0.82$, unconditionally; and for all depths $w \le 0.98$ modulo one isolated cell-crowding cap whose ledger constant is interval-certified ($\sup S_2^{\mathrm{gen}} \le 0.98465 < 1$ on $(0, 0.156]^2$, per-cell rigorous enclosures). On the deepest sliver $w \in (0.98, 1]$ the ledger inequality fails (interval-certified $S_2^{\mathrm{gen}} \ge 1.01405 > 1$ at $d = d' = 0.159$); there the targeted crowd-plus-sea attacks — 8 to 65 pairs, one per cell tranche, plus greedy mark-2 seas up to 130 atoms, re-run at $d = 0.158$ and 0.159 — never bring $F_1 - T$ below $+17.46$, against a floor requirement of 0, and the unconditional 8/9 backstop applies.*

Proof architecture (full details in the companion note). Partition the circle into 65 kernel-zero cells; per cell, the exploitable dip mass $\kappa_k(y) = \sup_{\text{cell}}[-R_y]_+$ sums to the capacity curve $\nu(y)$, with the exact curvature identity $\alpha := \sum_j u_j(2\pi j/N)^2 = \sum_k \psi'(g_k)^2 = 3.392677$ controlling the shallow regime ($\nu(y) = \alpha y^2(1 + o(1))$). Integer atoms harvesting a cell's dip pay in-cell atom-atom crosses ($\geq \psi(\text{dip-zone diameter})^2$ each) and their own integrality slack, capping the harvest at $8m_p\kappa_k$ per cell for marks ≤ 2 (with a certified stacking correction beyond); for pair depths $w \leq 1$ the closing inequality $8\nu(y) \leq 2\bar{a}(2y)^2$ follows in closed form from one harmonic-Taylor bound with the exact α and one certified finite constant (1.2% closing margin at $w = 1$); the deep probes are covered by the certified capacity ratios of Theorem 4.6. Pair-pair couplings are priced through the joint kernel $\Phi(d, d'; x) = R_{d+d'}(x) + R_{|d-d'|}(x)$ (never the two parts separately), writing $\nu_{\text{joint}}(d, d') = \sum_k \sup_{\text{cell}}[-\Phi]_+$ for its capacity curve and $\Phi_0 = \inf\{R_s(x) + R_t(x) : |x| \leq 32/65,\ s \leq 1/\pi,\ t \leq 1/(2\pi)\}$ for the in-cell coupling a second pair must pay; the ledger constant is then $S_2^{\text{gen}}(d, d') = [8\nu(d) + 4\nu_{\text{joint}}(d, d')]/(2\bar{a}(2d)^2)$ and the closure needs $S_2^{\text{gen}} \leq 1$. This carries a one-pair-per-cell crowding cap that is self-consistent for $w \leq 0.82$ (both sides interval-certified: $2 \max \nu_{\text{joint}} \leq 0.60910 < 0.64371 \leq \Phi_0$) and whose ledger constant beyond it is interval-certified to $w \leq 0.98$; beyond $w = 0.98$ the ledger fails (certified) and the closure defers to numerics plus the backstop. Deep single pairs ($d \geq 1$) fall to the crude bound $\bar{a}(2d)/\bar{a}(d) \geq 12.158$ nondecreasing. QED (architecture)

Theorem 4.8 (equal-depth pair-only configurations, exact, all depths)**.** *If c consists only of pairs, all at one common depth d (any positions, any integer multiplicities), then $F_1 \geq S_2 + 2\sum_p m_p^2$ — (MI) with a full doubling-cushion surplus at every depth.*

Proof. $c_s = 2\cosh(2\pi sd/N)\,\varphi(s)$ with φ the position-multiset form factor, and $4\cosh^2 = 2 + 2\cosh(\text{double})$: F_1 splits as twice the atom-only Frobenius of the position multiset ($\geq 2\sum_p m_p^2$ by diagonal positivity) plus a termwise-cosh-weighted copy ($\geq$ the same floor). □

Theorem 4.9 (unconditional spectral backstop, no restrictions)**.** *For every admissible configuration — any number of pairs, any depths, any marks — $N_d \geq (4/9)(3M - F_1)$. Hence for any law with mass N and $E[F_1] \leq (4/3)N(1 + \varepsilon)$:*

$$\frac{E[N_d]}{N} \;\geq\; \frac{8}{9}\Big(\frac{5}{6} - \frac{2}{3}\varepsilon\Big) = \frac{20}{27} - \frac{16}{27}\varepsilon = 0.7407\ldots - 0.593\,\varepsilon\,,$$

so the maximal conceivable pair advantage on the corner, in any regime whatever, is $\leq 5/6 - 20/27 = 5/54 = 0.0926$, unconditionally.

Proof. Let B be the Hermitian frequency matrix $B[j, j'] = \sqrt{u_j u_{j'}}\, c_{j-j'}$ (65×65): exactly $\operatorname{tr} B = M$ and $\operatorname{tr} B^2 = F_1$. B decomposes as $\sum_{\text{atoms}} m_a v_a v_a^\dagger + \sum_{\text{pairs}} B_p$ with each atom block PSD of rank 1. Each pair block is $B_p = m(pq^\dagger + qp^\dagger)$, with $p_j = \sqrt{u_j}\, e^{-2\pi i j\theta/N} e^{2\pi j d/N}$ and $q_j = \sqrt{u_j}\, e^{-2\pi i j\theta/N} e^{-2\pi j d/N}$ (the two summands of the conjugate-pair form factor of Section 2.3); on $\operatorname{span}\{p, q\}$ its nonzero eigenvalues are $m(\operatorname{Re}\langle q, p\rangle \pm |p||q|)$, and Cauchy–Schwarz gives $\operatorname{Re}\langle q, p\rangle - |p||q| \leq 0$, so $n_+(B_p) \leq 1$ — the finite-circle counterpart of the signature-$(1,1)$ block of Proposition 3.2(iv). By subadditivity of positive inertia, $n_+(B) \leq \#\text{atoms} + \#\text{pairs} \leq N_d$. For every real t, $3t - t^2 \leq (9/4)\mathbf{1}_{t>0}$; summing over the spectrum, $3M - F_1 = \sum_i (3\lambda_i - \lambda_i^2) \leq (9/4)n_+(B) \leq (9/4)N_d$. □

Remark. The 9/4 (rather than 2) is the price of a purely spectral argument — $3t - t^2$ exceeds 2 on $(1, 2)$ with max 9/4 at $t = 3/2$ — the same structural reason the inertia route of [AF26] cannot reach 5/6 by itself. The bound is, in compensation, depth-uniform, mark-uniform and interference-proof.

Status and the sharp conjecture. Roughly 10^4 adversarially optimized configurations (dip attacks, pair crowds, half-gap frustrated twins, deep seas, random mixed stress; seeds fixed, independent assembly) never brought F_1-T below $+2.000000$, and the minimum of $F_1-T-\sum_p 2(2m_p^2-3m_p+2)$ over every family is 0.000000, attained only in the depth $\to 0$ grid limit. The searches establish that the binding adversary is always shallow; that coordinated dip attacks can consume nearly the whole depth-dependent excess $2m^2(\bar{a}(2d)^2-1)$ (within 3% at $d=0.1$) but never any of the depth-independent doubling cushion $2m^2$ — exactly the split the capacity lemmas predict. We record the sharp conjecture (numerically exhaustive, not proved): $F_1 \geq \sum_a(3m_a-2)+4\sum_p m_p^2$ — every pair pays its full flattened-double cost and interference recovers nothing. Its near-tight witnesses ($+0.03$) show any proof must be capacity-sharp, which is why the theorems above target (MI), with its floor headroom of 2 per pair, instead.

Consequence. The model-level corner theorem extends from the atom-only class to the bulk of the admissible class of Section 2.4: single-pair columns at every admissible depth ($w \leq 2.8$), multi-pair columns at $w \leq 0.82$ unconditionally and $w \leq 0.98$ modulo the crowding cap (constant interval-certified), equal-depth pair-only columns at every depth — for both benchmarks, via (MI). The remaining admissible regimes — multi-pair columns at the deep-probe depths $w \in \{1.5, 2\}$ (inside the admissible class but beyond the theorems' range), multi-pair columns with a depth in $w \in (0.98, 1]$ (ledger failure certified there; coverage is the re-run attacks plus the backstop), and mass-unbounded crowds (not LP-relevant, since columns have mass N and deeper objects are garnish priced by Section 7's ladder) — are floored at 8/9 of the corner unconditionally and covered by converged pricing (O2); the $(0.98, 1]$ sliver's coverage is the re-run attacks under (O1) plus the backstop. Class-level exactness of $\delta_0 = 0$ is therefore proved on the closure's regimes and heuristic (pricing-backed, backstopped at 20/27) on that residue. A pair-assisted violation would have registered as a strictly positive marginal value, to be cut by pricing rather than by absorption; none is present on the closure's regimes. The residual coverage is stated in Section 8.3.

4.4 The computational certification: 940 records, the ε law, and the witness

The decision LP (column generation over explicit configurations, analytic-gradient multi-start pricing, exact-LP equality on the accumulated dictionary, final solutions re-verified in `Fraction` arithmetic on 10^{-12} rationalizations) was run over the full decision grid. The outcome at every record is absorption; the records are stored in machine-readable form.

The grid. $N = 64$: {matched, asymptotic} budgets $\times$ $W \in \{2,3,4,6,8\} \times C_{\text{led}} \in \{10, 100, 1000\}$ $\times$ $\varepsilon \in \{0.02, 0.05, 0.10\}$ $\times$ fuzz {coupled, none, $\Gamma \in \{0.05, 0.10, 0.25\}$} $\times$ budget setting {centered, set adversary-favorably within the error bars} = 900 records. $N = 128$: {matched, asymptotic} $\times$ W $\times$ fuzz {coupled, none} $\times$ the same two budget settings = 40 records. Total 940 records; every record solved to status success; LP duality gap $\sim 10^{-9}$.

The result. $\delta_0 = P_{\text{full}} - P_{\text{base}} = 0$ at every record: $\max|\delta_0| = 6.39 \times 10^{-13}$ over the 900 $N = 64$ records and 8.94×10^{-13} over the 40 $N = 128$ records, so $|\delta_0| < 10^{-12}$ at all 940; the maximum over the full grid is 8.94×10^{-13}, attained at $N = 128$. Per-W values at the primary point: -3.77×10^{-13} ($W = 2$), $+1.44 \times 10^{-13}$ ($W = 3$), 0.0 exactly ($W = 4, 6, 8$); the fit $\delta_0(W) = \delta_\infty + cW^{-\kappa}$ degenerates to $\delta_\infty = 0$ with zero residual, and $d\delta_0/d\Gamma = 0$. The optima are not merely equal but achieved by identical optimal laws: the base-optimal law itself satisfies the entire cubic block.

The exact ε law. At asymptotic budgets the joint optimum obeys $P(\varepsilon) = 5/6 - (2/3)\varepsilon$ — the grid-decoupled corner of Theorem 4.3 — at every tested ε:

ε	P (asymptotic, LP)	$5/6 - (2/3)\varepsilon$	$\lvert\delta_0\rvert$
0.02	0.8200000000002683	0.82	2.8×10^{-13}
0.01	0.8266666666669427	0.8266666...	2.4×10^{-13}
0.005	0.8300000000002841	0.83	2.2×10^{-13}
0.002	0.8320000000003910	0.832	1.0×10^{-13}

Matched-variant values differ only through the finite-size budget center (e.g. $P = 0.8368627361$ at $\varepsilon = 0.002$, from $B_1 = 84.712 < (4/3)\,64 = 85.33$). Moreover, adjoining the λ'-Frobenius budget row alone to the two-moment baseline leaves the optimum at P_{base} at every record: that row by itself adds nothing.

Primary decision point (flat/flat, $\lambda' = 1/2$, $N = 64$, matched budgets, $\varepsilon = 0.05$, $C_{\text{led}} = 100$, coupled fuzz): $P_{\text{full}} = P_{\text{base}} = 0.8050956815843511$, $\delta_0 = -3.8 \times 10^{-13}$, identical optimal laws; the same P at fuzz = none (the fuzz row is not load-bearing for the absorption). At $N = 128$, $P_{\text{full}} = P_{\text{base}} = 0.8024488905983833$ (matched) and 0.8000000000004773 (asymptotic) at $\varepsilon = 0.05$, with $\delta_0 = 0$ at all 40 records.

The witness (`witness_N64.json`; $N = 64$, matched budgets, $\varepsilon = 0.05$, fuzz = none). Three columns, all supported on the 65-site ψ_1-zero grid (spacing $64/65 = 0.9846153846$), marks $\{1, 2\}$ only, no pairs, $n_- = 0$:

weight	column	N_d	F_1 ($= \sum m^2$, exact)	F'	C'
0.5299906673474316	vacancy lattice: 64 simples on 64 of the 65 sites	64	64	125.0909091 = 4128/33	245.4508724
0.16040139164388625	16 doubles + 32 simples on an explicit 48-site subset	48	96	153.7604346	411.0865130
0.30960794100868216	32 doubles on an explicit 32-site subset	32	128	135.1959163	301.3541285

Row values in this table are the clean-integer-grid values (`clean_checks[].*_clean` — the canonical hand-checkable description); the optimizer's stored law rows (`law[].F1/Fp/Cp`) differ from them by at most 4.1×10^{-5} per row (per column $3.4 \times 10^{-6}/1.6 \times 10^{-5}/4.1 \times 10^{-5}$), and the aggregates below are the stored law's, which saturate the budget edges exactly. Aggregates: $E[F_1] = 88.94775275723633$ = the upper Frobenius edge $B_1(1 + \varepsilon)$ exactly; $E[F'] = 132.818$ (strictly inside its band $[128.34, 141.85]$); $E[C'] = 289.32716513260516$ = the lower cubic edge $B_3(1 - \varepsilon)$ exactly (the doubles corner is cubic-short, per $G(1/2) > 0$, and the arrangement supplies the budget from below); $E[N_d]/N = 51.52612362142/64 = 0.8050956815846875 \le 5/6 + 0.01$ (the absorption bar of Section 2.5). Active rows at the optimum: only the upper F_1 budget edge and the lower cubic edge; every ladder row slack by orders of magnitude — $E[n(2)] = 3.50$ and $E[n(3)] = 0.94$ against caps $C_{\text{led}} N V^{-4} = 400$ and 79.0, all higher rows zero, and the vacancy column's $n(V)$ vector identically zero. Every row was re-verified in exact `Fraction` arithmetic (all 10 checks pass, objective deviation 5×10^{-13}), and all three columns were recomputed independently through a position-space path (reproduction to 10^{-6} or better, feasibility re-confirmed at $C_{\text{led}} = 100$ and $C_{\text{led}} = 10$).

The mechanism is explicit: $F_1 = \sum m^2$ exactly on all three columns (Theorem 3.9 — the equality case of Lemma 4.1), so the bandwidth-one reading transfers the `lemmaR_tight` corner verbatim, while the sub-grid arrangement (which sites, which clusters) freely meets the λ' rows (Proposition 3.10): the two doubles-subsets' occupancy-2 clustering supplies the cubic cross-terms ($C' = 411.09$ and 301.35 against isolated-block values 160 and 256) at zero cost in N_d and zero cost at $\lambda = 1$. The witness uses no pairs, no marks ≥ 3, no spectral escape — arrangement alone.

The role of pricing. No improving column (reduced cost $< -10^{-6}$) was found in $S = 200$ pricing restarts at the primary and tightest-asymptotic points ($S = 100$ at $N = 128$, recorded but with no stored verification block; $S = 40$ at W-scan re-checks; $\min_{\text{rc}} = 0.0$, 0 improving columns at every stored verification). The absorption conclusion does not rest on pricing optimality: the witness is exhibited and verified, and $P_{\text{full}} = P_{\text{base}}$ is an exact LP equality on the accumulated dictionary in any case. Pricing convergence supports only the claim that no configuration class outside the dictionary would carry a strictly positive marginal value (and more columns can only lower both optima together, the anti-absorption direction).

Robustness. The conclusion was stress-tested along eight independent axes: recomputation of the witness; a per-column ladder at $C_{\text{led}} = 4$, barely above the null floor 3.82; a missing-row hunt; near-CUE pinning; window-degeneracy probes; the pair-channel attack; budget-center stress across ε in 0.002–0.10 and Γ to 0.25; and seed and scale stability. Absorption persists along all eight. Two of the axes produced further results — on the near-CUE class and the p_1 objective — developed in Sections 5 and 6.

5 The near-CUE class: absorption inside the unconditional data class

The adversary of Section 4 is free to hold anti-correlated (non-CUE) bandwidth-one data. The strongest form of the no-go pins the adversary inside the near-CUE class — the data class of the PairCeiling formalization accompanying [AF26] — and asks whether the cubic block has positive marginal value there. It does not, and the mechanism is structural: at the pinned optimum the cubic row is slack (Section 5.3).

5.1 The licensing theorem, and its three riders

The pinning rows $\big|E_w|c_j|^2 - j\big| \leq \tau_2$ for $1 \leq j < N$ at $\lambda = 1$ (edge row free) are licensed by Theorem 1 of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [BGSTB24] (arXiv:2306.04799 v1, June 2023; Acta Arithmetica, 2024), which is unconditional and pointwise in α, uniform on the closed band $0 \leq \alpha \leq 1$:

$$F(\alpha) = T^{-2\alpha}(\log T + O(1)) + \alpha + O\big(1/\sqrt{\log T}\big),$$

for the form factor defined with zeros counted with multiplicity and off-line zeros entering with the depth weight $x^{\delta+\delta'}$ — exactly the cosh depth-weighting the model's form factor c_j applies. With $\alpha = j/N$, each open-band row is licensed pointwise per row, unconditionally, in the depth-weighted multiplicity-counted form (directly, or through their Lemma 5 admissible kernel class: even, L^1, supported in $[-1,1]$, Lipschitz at 0 — the bandwidth-one class); the diagonal spike is confined to the edge rows the pinning class above leaves free, and the Poisson factor acts below the model's $1/N$ bin resolution whenever $N \ll \log T$. The uniformity is on the closed band $0 \le \alpha \le 1$, not $|\alpha| < 1$: their Theorem 1 incorporates an earlier improvement, which they cite, extending the range "up to $\alpha = 1$ with explicit error terms".

The three licensing riders (binding on every use of this section's results):

1. **Depth-aggregation only.** The licensed statistic never separates on-line from off-line mass: the Remark after their Theorem 2 states the method "neither requires nor provides any information" about whether the zeros are on the critical line. What is pinned is the cosh-weighted aggregate $|c_j|^2$ (pairs entering with depth weights), never an on-line-only statistic. Shedding the far-off-line coupling term costs an extra hypothesis — the thin-box hypothesis or the strong zero-density hypothesis, which their Theorems 2–3 (the 61.7% simple-zeros results) consume; the row class used here needs none of that.
2. **Finite-T slack $O(N/\sqrt{\log T})$.** The licensed per-row slack at finite T is $O(N/\sqrt{\log T})$ grid units (plus the edge spike), so $\tau_2 \in \{1, 4\}$ is the fixed-N, $T \to \infty$ asymptotic reading — legitimate for the model's asymptotic semantics, but at finite T the $O(1)$-unit pinning overstates the theorem until $\log T \gg N^2$.
3. **Second-moment scope.** The licensing theorem [BGSTB24] stops at the second moment — pair correlation only; nothing in it licenses the cubic budget column, whose $c_3(\lambda')$ identification remains open (the cubic budget center, Section 8.3).

In sum: the pinning rows are an unconditional asymptotic data class, pointwise on the open band in the depth-weighted form, and the *strength* of the near-CUE conclusion — absorption inside the two-moment certificate's own data class — inherits riders 1–3.

5.2 $\delta_0' = 0$ exactly, both N, both centerings

Row class: all the rows of Section 2.5 (fuzz = none — the harsher variant for absorption) plus the pinning rows at τ_2. Four cells at $N = 64$ (matched budgets, $C_{\text{led}} = 100$, $W \le 8$):

τ_2	ε	$P_{\text{full}}^{\text{pin}}$	$P_{\text{base}}^{\text{pin}}$	δ_0'	reading
1	0.05	0.8332334059839699	0.8332334059839699	**0.0** (exact)	absorption (exact)
1	0.02	0.8359542395410772	0.8359374423865479	1.68×10^{-5}	absorb (0 within the residual band $[-3.1 \times 10^{-5}, +1.8 \times 10^{-5}]$)
4	0.05	0.8150864520465212	0.8137831067863290	1.30×10^{-3}	bounded residual, unconverged upper bound
4	0.02	0.8316328692004420	0.8301979402692543	1.43×10^{-3}	bounded residual, unconverged upper bound

At the primary cell ($\tau_2 = 1$, $\varepsilon = 0.05$), $\delta_0' = 0$ is an exact LP equality — identical optima — at every one of 9 cumulative $S = 200$ verification passes across a ∼1,100-column dictionary enrichment; the common level fell $0.8333622 \to 0.8332334$ while the equality never broke (residual improvements are common-mode), and the reduced-cost bound for the true marginal value is $|\delta_0'| \le 1.2 \times 10^{-5}$. Stability under budget re-centering: the asymptotic variant gives $P_{\text{full}} = P_{\text{base}} = 0.8332332347490976$, $\delta_0' = 0.0$ exact.

$N = 128$ **confirmation.** The pinned LP was infeasible on the raw dictionary and was bootstrapped with 300 CUE and doubles-decorated-CUE columns (anti-absorption-safe), then built out over 10 cumulative $S = 200$ passes to 6,457 columns at the matched cell (6,739 after the asymptotic re-centering): $P_{\text{full}} = P_{\text{base}} = 0.8374218534574983$ with $\delta_0' = 0.0$ exact at the matched cell, and 0.8373740026259704 with $\delta_0' = 0.0$ exact at the asymptotic re-centering. Build-out invariance: $\delta_0' \leq 8 \times 10^{-16}$ at every recorded dictionary re-solve (1,842/6,457/6,739 columns) — the conclusion is independent of the convergence level. Level trend: $P^{\text{pin}} = 0.8374$ ($N = 128$) vs 0.8332 ($N = 64$), both $\sim 5/6 - O(10^{-3})$, the pinned-baseline finite-size trend.

5.3 Absorption by slack

The near-CUE witnesses explain the mechanism. At $N = 64$: a 64-column law, marks $\{1, 2\}$ only, no pairs, no marks ≥ 3, with $E|c_j|^2 = j+1$ for every $j = 1, \ldots, 63$ to 2×10^{-12} (maximum deviation 1.84×10^{-12}, at $j = 62$) — all 63 upper pinning rows active at exactly the $+\tau_2$ allowance — and the only active rows at the pinned optimum are the 63 pinning rows (and mass): F_1, F', C', and every ladder row are strictly interior. The cubic block does not bind. The $+\tau_2$ allowance pays for the doubles; the cubic budget is met with room to spare. `Fraction` re-verification passes on the full row class (worst pinning margin -1.0×10^{-9}, inside the documented 10^{-6} grace). At $N = 128$ the same structure: 128 columns, marks $\{1, 2\}$, no pairs, upper pinning edge ridden at $\tau_2 - 10^{-5}$ (an interior shrink absorbing a solver-tolerance overshoot; the `Fraction` re-verification against the true $\tau_2 = 1$ box passes with margin $+1.0 \times 10^{-5}$ strictly interior), $E[n_{\text{simple}}]/N = 0.6748$.

This is a stronger statement than absorption in the unpinned class: inside the near-CUE class the cubic row is not absorbed by an adversary straining against it — it is slack at the optimum. There is no escape route inside the certificate's own data class, and the cubic block has no positive marginal value there, at both lattice sizes, at both budget centerings, subject to riders 1–3 of Section 5.1.

5.4 The $\tau_2 = 4$ annotation

At loose pinning ($\tau_2 = 4$) the cubic row is active and the residuals $\delta_0' = 1.30 \times 10^{-3}/1.43 \times 10^{-3}$ ($\varepsilon = 0.05/0.02$) exceed their pricing bands ($[0.9 \times 10^{-3}, 1.5 \times 10^{-3}]$) and were still declining under enrichment ($1.28 \times 10^{-3} \to 1.17 \times 10^{-3}$ over five passes): they are unconverged upper bounds at loose pinning, an order of magnitude below the 10^{-2}–3×10^{-2} δ_0 the proposal projected, obtained from shortened runs and reported as such. They are diagnostic annotations, not certificate claims, and they sharpen the scoping: whatever residual cubic signal exists at this scale lives strictly between the near-CUE class ($\tau_2 = 1$: exact zero by slack) and the free unpinned class (exact zero by witness), i.e. in loosely pinned intermediate worlds, at magnitudes far below the scale at which the question was posed.

6 The simple-fraction benchmark

The headline of the two-moment certificate [AF26] is the simple-zero fraction (2/3 at flat window; 0.6725 window-optimized; ceiling 0.6818287), so a no-go scoped only to N_d would be incomplete. The computation was therefore re-run with the objective $\min E_w[p_1]$, $p_1(c) = (\#\text{mark-1 atoms} + 2 \times \#\text{mult-1 pairs})/N$, on the dictionary as enriched by the near-CUE runs of Section 5 (5,134 columns), with the stopping criterion — no improving column in $S = 200$ pricing restarts — applied to both systems:

cell	$p_{1,\text{full}}$	$p_{1,\text{base}}$	δ_{p_1}	analytic corner $2-(B_1/N)(1+\varepsilon)$	converged
matched, $\varepsilon = 0.05$	0.6101913631697143	0.6101913631684138	1.3×10^{-12}	0.6101913631681821	yes: $S = 200$ clean, $\min_{\text{rc}} = 0.0$, both systems
asymptotic, $\varepsilon = 0.002$	0.6640000000012899	0.6640000000004871	8.0×10^{-13}	$0.6640000000000001 = 2-(4/3)(1.002)$	yes: $S = 200$ clean, both systems

These are the only cells reported here in which the pricing stopped clean for both systems at once: zero improving columns at $S = 200$ for the full and base systems. Cross-checks on the final dictionary: coupled fuzz at the primary cell $\delta_{p_1} = 2.9 \times 10^{-13}$; W-scan over $\{2, 3, 4, 6, 8\}$ max $|\delta_{p_1}| = 1.3 \times 10^{-12}$ (at $W = 8$; all $< 1.4 \times 10^{-12}$); the labeled on-line-only diagnostic variant 3.3×10^{-13}. The optimum is the exact marks-$\{1, 2\}$ doubles corner of Theorem 4.3(b) — pairs cannot help (a mult-1 pair buys 2 simple points at $\geq$ twice the two-simples Frobenius cost; a mult-2 pair is dominated by two doubles) — and the $\varepsilon \to 0$ value is $2 - 4/3 = \mathbf{2/3}$, the model's bandwidth-one analog of Montgomery's simple-zeros corner. The witness laws are the same ψ_1-grid family as the witness of Section 4.4, reweighted per cell, rational verification passing.

Two scoping notes. (i) The cited 0.6818287 ceiling belongs to the *window-optimized* certificate class (the Theorem-D effect), which the flat/flat row system used here does not consume; the flat-window model corner is 2/3, and what is proved here is that the cubic block does not move the flat/flat optimum for either objective, leaving the 0.6818287 ceiling untouched. (ii) With this section, the no-go covers both benchmarks: the distinct-zeros corner 5/6 and the simple-fraction corner 2/3, each with $\delta = 0$ at convergence.

7 Capacity and vacuity: no spectral escape

Two theorems close the spectral-escape routes from opposite ends, and together with the decision program they complete the pricing: a scalar cubic tail row with any divergent cutoff is unconditionally vacuous (Theorem 7.4 — the half of the no-go that needs no linear program at all); and under the all-V ladder the same channel is capped at $o(N)$ with a sharp constant (Theorem 7.1). What remains is position/interference freedom — which is exactly the channel the witness of Section 4.4 uses.

7.1 The sharp capacity theorem

Setting (the continuum garnish LP). Heights normalized so the ladder's absolute floor is 1. Spectral escape is a positive measure μ on $[1, \infty)$ — $\mu([h, \infty))$ is the count per N of escaped eigenvalue mass at height $\geq h$ — subject to

$$\text{(Frobenius slack)} \quad \int h^2 \, d\mu(h) \leq \varepsilon_g \,,$$
$$\text{(all-}V\text{ ladder)} \quad \mu([h, \infty)) \leq C h^{-4} \quad \text{for every } h \geq 1 \qquad (C = C_{\text{led}}),$$

and the objective is the absorbable cubic charge $\int h^3 \, d\mu(h)$. (Equivalently, in tail-function form $N(h) = \mu([h, \infty))$: maximize $N(1) + 3 \int_1^\infty h^2 N \, dh$ subject to $N(1) + 2 \int_1^\infty h N \, dh \leq \varepsilon_g$ and $N \leq C h^{-4}$; the forms are identical by integration by parts.)

Theorem 7.1 (sharp capacity)**.** *For every $C > 0$ and $0 < \varepsilon_g \le 2C$:*

$$\max_{\mu} \int h^3 \, d\mu = 2\sqrt{2}\,\sqrt{C\varepsilon_g}\,,$$

attained by the min-profile $N_*(h) = C\min(a^{-4}, h^{-4})$, $a = \sqrt{2C/\varepsilon_g}$ (as a measure: no mass on $[1, a)$; density $4Ch^{-5}\,dh$ on (a, ∞)).

Proof. Upper bound: for any feasible μ and any $a > 0$, the pointwise inequality $h^3 \le ah^2 + (h^3 - ah^2)_+$ on $[1, \infty)$ gives

$$\int h^3 \, d\mu \le a\varepsilon_g + \int (h^3 - ah^2)_+ \, d\mu\,.$$

Writing $g(h) = (h^3 - ah^2)_+ = \int_a^h g'(s)\,ds$ for $h \ge a$, with $g'(s) = 3s^2 - 2as > 0$ on (a, ∞), Tonelli and the ladder give

$$\int g \, d\mu = \int_a^\infty g'(s)\,\mu([s,\infty))\,ds \le C\int_a^\infty (3s^2 - 2as)s^{-4}\,ds = C\Big(\frac{3}{a} - \frac{a}{a^2}\Big) = \frac{2C}{a}\,.$$

So $\int h^3\,d\mu \le a\varepsilon_g + 2C/a$ for every $a > 0$; minimizing over a (minimum at $a = \sqrt{2C/\varepsilon_g}$) gives $2\sqrt{2C\varepsilon_g}$. Attainment: $\varepsilon_g \le 2C$ makes $a \ge 1$, so N_* is an admissible tail function; its Frobenius charge is $Ca^{-4} + Ca^{-4}(a^2 - 1) + Ca^{-2} = 2C/a^2 = \varepsilon_g$ exactly, and its cubic charge is $C/a + 3C/a = 4C/a = 2\sqrt{2}\sqrt{C\varepsilon_g}$, meeting the bound. ☐

Remark (the two-sided squeeze, made exact)*.* The optimal dual split is

$$[a \times \text{Frobenius}] + [\text{ladder integral above } a]\,:$$

bounded heights are blocked by the Frobenius cost (mass at height h buys h^3 of cubic but pays h^2 of Frobenius, ratio $h \le a$), divergent heights by the ladder cap. The absolute floor C_{abs} (normalized to 1) appears nowhere in the upper bound — only in the attainability condition $a \ge 1$ — which is why a criterion of the form "the marginal value is positive iff $2C_{\text{led}}/C_{\text{abs}} < 4/3$" cannot decide the question: sharp constants at the floor are irrelevant against this adversary class.

Proposition 7.2 (the naive profile is infeasible)**.** *The point-mass-plus-tail profile (a ladder-saturating atom of size $n_0 = Cb^{-4}$ at height $b = \sqrt{3C/\varepsilon_g}$, plus the ladder-saturating tail above b) has Frobenius charge exactly ε_g and cubic charge $5C/b = (5/\sqrt{3})\sqrt{C\varepsilon_g} = 2.8868\sqrt{C\varepsilon_g}$ — but it violates the cumulative ladder on the open interval $(2^{-1/4}b, b)$, where its count $2Cb^{-4}$ exceeds the cap Ch^{-4}. The true optimum is Theorem 7.1's $2\sqrt{2} = 2.8284271247\ldots < 5/\sqrt{3}$; the earlier estimates $5/\sqrt{3}$ and the two-term dyadic $2\sqrt{C\varepsilon}$ are of the same order and are superseded. The constant is confirmed three independent ways: the min-profile derivation above, an analytic re-derivation together with a linear program (2.8244 at $h_{\max} = 400$, gap = the predicted truncation tail $3/h_{\max}$), and a separate numerical optimum.*

Proof. The tail part has density $4Ch^{-5}$ on (b, ∞). Frobenius: atom Cb^{-2} + tail $2Cb^{-2} = 3Cb^{-2} = \varepsilon_g$ at the stated b. Cubic: atom C/b + tail $4C/b$. Ladder violation: for $2^{-1/4}b < h < b$ the cumulative count is $n_0 + Cb^{-4} = 2Cb^{-4}$ while the cap is $Ch^{-4} < 2Cb^{-4}$ throughout the open interval. ☐

Corollary 7.3 (capacity is $o(N)$: the positive complement of the no-go)**.** *In the model's normalized units, the maximal cubic shift purchasable by spectral escape under the row system with the all-V ladder in force is*

$$|\Delta \mathit{cubic}| \le 2\sqrt{2}\,\sqrt{C_{\text{led}}\varepsilon_g}\,N = o(N) \qquad \textit{as } \varepsilon_g \to 0 \textit{ at fixed } C_{\text{led}}$$

(e.g. $C_{\text{led}} = 1000$, $\varepsilon_g = 10^{-8}$: capacity $0.0089N$). A doubles-vs-pairs-scale swing ($\Theta(N)$, e.g. $(4/3)N$) can never be produced by spectral escape once the all-V ladder is consumed: any absorption of the cubic block must run through position/interference freedom — the channel Theorem 3.9 exhibits and the witness uses. (In the decision LP this theorem enters as the coupled fuzz rows, tangent-cut on the concave $\sqrt{\ }$; the fuzz row is not binding at the optimum.)

Numerical confirmation (`verify_t5.py`, with results in `verify_t5_out.json`): attainment profile exact to $4 \times 10^{-11}/9 \times 10^{-13}$; an independent discretized LP over point masses converges to 0.4000 from below under joint grid-and-truncation refinement — tail-corrected values 0.3950/0.3985/0.3991 at (600 points, $h_{\max} = 2000$), (2000 points, $h_{\max} = 2000$), (4000 points, $h_{\max} = 8000$), $C = 1$, $\varepsilon_g = 0.02$, each the raw LP value plus the predicted truncation tail $3C/h_{\max}$ — never exceeding the bound; the naive profile's value 0.408248 and its ladder violation are reproduced; duality and scaling verified across (C, ε) over five orders of magnitude.

7.2 Divergent-cutoff vacuity: the garnish theorem

Setting (the row system in the divergent-cutoff form the proposal stated; family semantics). $N = N(T) \to \infty$; $V_0 = V_0(T) \to \infty$ any divergent cutoff with $V_0 = o(N^{1/3})$ (any polylog cutoff qualifies — in particular the proposal's $(\log\log T)^3$ and every $(\log T)^A$). The as-written system consumed its equalities "with $o(N)$ precision" — no fixed tolerance rate — so feasibility is a property of families (c_T) of model configurations (no mark cap was imposed as written), with spectral data read from isolated-block spectra, admissible iff

- (W1) mass: $\sum m = N$ (exact);
- (W2) Frobenius: $F = \kappa N + o(N)$, $\kappa = \kappa(1) = 4/3$ (the as-written system's bandwidth-one Frobenius row);
- (W3) signed cubic: $C = c_3 N + o(N)$;
- (W4) scalar tail row: $\sum_{|\lambda_i| \geq V_0} |\lambda_i|^3 = o(N)$;
- (W5) count/inertia rows imposed only at thresholds $V \geq V_0$, plus n_- at most the number of pairs;

objective $\liminf N_d/N$ (or $\liminf p_1$). (A fixed tolerance rate shrinking faster than $1/V_0$ would be a stronger system than the one written: the consumed prime-side equalities carry no such rate, so imposing one would assume a stronger input than the sources provide.)

Theorem 7.4 (garnish absorption)**.** *Let (c_T) satisfy (W1), (W2), (W4), (W5), contain at least $s_0 N$ simple on-line atoms for a fixed $s_0 > 0$ (a simple-atom fraction, unrelated to the marginal value δ_0), and fall short of the (W3) cubic equality by $\Delta_T N$ with $0 \leq \Delta_T \leq \Delta_{\max}$ fixed. Put $h_0 = \lfloor V_0/2 \rfloor$, $n_0 = \lceil \Delta_T N/h_0^3 \rceil$, and let c'_T be c_T with n_0 additional isolated on-line atoms of mark h_0 (at unoccupied, well-separated positions) and $n_0 h_0$ simple atoms deleted. Then for all large T, c'_T satisfies every row of the as-written system, and*

$$N_d(c'_T) = N_d(c_T) - n_0(h_0 - 1) \leq N_d(c_T), \qquad p_1(c'_T) \leq p_1(c_T).$$

Row-by-row (per N; every shift $o(1)$): mass exact 0; Frobenius $+\ \Delta_T/h_0 + O(1/h_0^2 + h_0^2/N) = o(1)$; cubic $+\ \Delta_T + o(1)$ (landing the (W3) equality); tail row $+\ 0$ (the garnish sits at $h_0 < V_0$); count rows at $V \geq V_0$: $+\ 0$; inertia unchanged (only positive on-line eigenvalues added).

Proof. Feasibility: $n_0 h_0 \le \Delta_{\max} N/h_0^2 + h_0 = o(N) < s_0 N$ for large T — enough simples exist to delete. Each garnish atom is an isolated on-line atom of mark h_0: eigenvalue h_0, Frobenius h_0^2, cubic h_0^3, mass h_0. Mass: adds and deletes $n_0 h_0$ exactly. Cubic: adds $n_0 h_0^3$ in $[\Delta_T N, \Delta_T N + h_0^3]$, subtracts $n_0 h_0 \le \Delta_{\max} N/h_0^2 + h_0$; net $\Delta_T N + O(h_0^3) + O(N/h_0^2) = \Delta_T N + o(N)$ since $h_0 \to \infty$ and $h_0^3 = o(N)$ — (W3) now holds. Frobenius: adds $n_0 h_0^2 \le \Delta_{\max} N/h_0 + h_0^2 = o(N)$, inside (W2)'s window. Tail and count rows: every garnish eigenvalue equals $h_0 = \lfloor V_0/2 \rfloor < V_0$, and all thresholds sit at $V \ge V_0 > h_0$, so the added spectrum is invisible; deletions only decrease those rows. Objectives: N_d changes by $+n_0 - n_0 h_0 = -n_0(h_0 - 1) \le 0$; the deleted atoms were simple and the added ones have mark $h_0 \ge 2$, so p_1 decreases. ☐

Remark (the growth cap is removable). The hypothesis $V_0 = o(N^{1/3})$ is a convenience, not a constraint: for an arbitrary divergent cutoff, cap the garnish height at $h_0 = \min(\lfloor V_0/2 \rfloor, \lfloor N^{1/4} \rfloor)$. Then $h_0 \to \infty$, $h_0^3 \le N^{3/4} = o(N)$, and $h_0 < V_0$ still holds, so every estimate in the proof goes through verbatim — the vacuity covers any divergent cutoff.

Corollary 7.5 (vacuity of the scalar divergent-cutoff cubic row)**.** *Suppose the two-moment system $\{$(W1), (W2)$\}$ admits optimizer families with a fixed positive simple fraction, spectra bounded by a fixed constant, and bounded cubic deficit ($C(c_T) \le c_3 N + o(N)$, deficit $\le \Delta_{\max} N$). Then, for both benchmarks, the infimum over the as-written system equals the infimum over the two-moment system: the marginal value of the entire block $\{$(W3) + (W4) + (W5)$\}$ is zero — $\delta_0 = 0$ as written. The deficit hypothesis is the actual situation: the corner families of Theorem 4.3 have marks in $\{1, 2\}$ (spectra bounded by 2, simple fraction 2/3 in the limit) and undershoot the cubic budget (Corollary 3.6: $G(1/2) = +1/2 > 0$ — the null budget sits strictly above the doubles plane, so doubles-saturated configurations are cubic-short; the witness accordingly sits at the lower cubic edge). Whatever cubic demand the row was hoped to impose on the two-moment-degenerate corner — up to any fixed $\Delta_{\max}$, including the full "$(4/3)N$ doubles-vs-pairs swing" of the proposal — the garnish supplies it invisibly, at an objective decrease.*

Proof. One direction is trivial (more rows can only raise the infimum). Conversely, take the corner families: bounded spectra plus $V_0 \to \infty$ make (W4) and the $V \ge V_0$ count rows identically zero for large T, so they satisfy (W1), (W2), (W4), (W5); apply Theorem 7.4 to land (W3); the garnished family is admissible for the whole system at a weakly smaller objective. ☐

Remark 7.6 (what this does and does not say).

1. **It kills the scalar row, not the underlying idea.** The vacuity is about the as-written system, whose count/tail information starts only at V_0. The proposal's own Chebyshev proof in fact yields the all-V ladder $n(V) \le C_{\mathrm{led}} N V^{-4}$ for every $V \ge C$ at $\vartheta < 1$ — more than the divergent-cutoff form in which it was stated; under that system the garnish channel is not free but capped, and the cap is exactly Theorem 7.1's $2\sqrt{2}\sqrt{C_{\mathrm{led}}\varepsilon_g}\, N = o(N)$. Claiming exactly what that proof yields is precisely the difference between Corollary 7.5 and Theorem 7.1.

2. **The garnish needs unbounded marks** ($h_0 = \lfloor V_0/2 \rfloor \to \infty$): it lives outside every fixed-W alphabet — which is why the adversary class here carries a divergent mark alphabet W, and why a violation found only at marks $\{1, 2\}$ would not settle the question. (A pair-based garnish with slowly divergent pair multiplicity reaches the same conclusion through Proposition 3.2's deep-pair blocks.)

3. **The three-part structure of the no-go.** Theorem 7.4/Corollary 7.5 dispose of spectral escape under the divergent-cutoff row; Theorem 7.1 caps it under the all-V ladder; and the computational content (Sections 4–6) is that the remaining channel — position/interference

freedom — absorbs the cubic block too, at $\delta_0 = 0$ exactly. Together these are the sharpened no-go.

Numerical confirmation (`verify_t6.py`): the shift table at $\Delta = 4/3$, $V_0 = (\log\log T)^3$ over a $\log T$ grid extended to 10^6 — Frobenius shift $4/(3h_0) = 0.0992$ down to $0.0010 \to 0$, N_d shift negative throughout, cubic shift $+4/3$ exactly, tail and count rows 0 — reproducing two independent computations of the same table; an abstract LP demonstration in which the cubic equality is infeasible without the garnish variable (which would otherwise read as a positive marginal value) and returns to the two-moment corner $5/6 - O(\text{tol})$ with it; and an exact-rational instance with all shifts exact.

8 Scope, residuals, and what survives

This section states where $\delta_0 = 0$ is exact, where it is a bounded residual, what is finite-scale, and what remains open, and closes with the one range of configurations in which the cubic row does carry a positive marginal value.

8.1 The residual table

The exact zero is flat/flat-specific. Theorem 3.9's mechanism says exactly why: the flat bandwidth-one kernel's zeros are equally spaced, so one site grid kills all $\lambda = 1$ cross-terms simultaneously; a cosine window's kernel zeros are not equally spaced, so no grid does, and the degeneracy breaks by a microscopic but nonzero amount. The complete residual map:

configuration	residual δ_0	status
flat/flat, entire 940-record decision grid	0 ($< 10^{-12}$)	exact; witness-backed
flat at $\lambda = 1$ / $\cos(1.6s)$ at λ'	5.95×10^{-14} (ε .05), -1.14×10^{-13} (ε .02)	exact-zero class: changing the λ' window moves nothing — the absorber's freedom is positional, not spectral
$\cos(\sqrt{2}\,s)$ [Montgomery–Taylor] at $\lambda = 1$ / flat	$+6.41 \times 10^{-5}$ (ε .05), $+2.28 \times 10^{-5}$ (ε .02)	bounded residual: the MT cosine breaks the exact grid degeneracy; unconverged upper bounds, three orders below the ε-slack scale
$\cos(\sqrt{2}\,s)$ [MT] / $\cos(1.6s)$	$+6.27 \times 10^{-5}$ (ε .05), $+1.91 \times 10^{-5}$ (ε .02)	same class
near-CUE $\tau_2 = 1$ (certified, $N = 64$ and 128, both centerings)	0 exact (reduced-cost bound $\leq 1.2 \times 10^{-5}$ at $N = 64$)	absorption by slack (Section 5.3)
near-CUE $\tau_2 = 4$ (loose box, diagnostic)	$1.30 \times 10^{-3}/1.43 \times 10^{-3}$, pricing bands $[0.9 \times 10^{-3}, 1.5 \times 10^{-3}]$	unconverged upper bounds, an order below the 10^{-2}–3×10^{-2} scale at which the question was posed; the cubic row is active here
independent tight-ε near-CUE probes (dictionary-limited, no column generation)	$\leq 5 \times 10^{-4}$, non-monotone in ε	upper bounds, same order as the documented $\sim 3 \times 10^{-4}$ pricing uncertainty — the regime where any residual signal would concentrate

Every nonzero entry is an unconverged upper bound on the converged marginal value, and every one sits at least an order of magnitude below the scale at which the question was posed. The scope of the exact-zero claim is therefore this: $\delta_0 = 0$ exactly at the primary flat/flat configuration; $\leq O(10^{-4})$ across the window family and $\leq O(5 \times 10^{-4})$ under near-CUE pinning at that scale, within pricing-convergence uncertainty; and at $\tau_2 = 1$ the near-CUE bound is replaced by the exact zeros of Section 5.

8.2 Finite scale

The computational certification is at $N = 64$ and $N = 128$, with the same conclusion at both (and across two budget centerings whose cubic centers differ by 5% — the F_1 centers by 0.7%, F' by 2.6%). It is not extrapolated beyond that: the continuum analog of the decoupling mechanism is exact (Proposition 3.11 — the integer lattice under the sinc kernel, with the finite-N grid its exact periodization), so the absorption has a structural reason to persist at every scale, and the model-level theorems of Sections 3, 4.1–4.3, and 7 are N-generic (the pair-channel capacity constants were certified at $N = 64$, the primary geometry; the $N = 128$ re-run reproduces every identity and inequality — see (O4) below). No claim about N beyond 128 rests on computation.

8.3 Open questions and residual coverage

- **The cubic budget center.** The identification $c_3(\lambda') = m_3(\lambda')$ ($= 5$ at $\lambda' = 1/2$) rests on Rudnick–Sarnak-range GUE 3-level correlations and on the polylog-window uniformity of the diagonal method of [AF26, §5]; Theorem 3.5 proves the sine value, not the identification. The absorption conclusion is insensitive to it: $\delta_0 = 0$ across ε in 0.002–0.10 and Γ to 0.25, and across budget centers differing by 5% — an $O(1)$-per-zero error in c_3 cannot flip it (one of the robustness checks recorded in Section 4.4). Nothing in [BGSTB24] licenses anything at third order (rider 3).

- **The ladder's provenance.** The all-V ladder consumed by Theorem 7.1's system rests on the all-V spectral-tail statement, whose step from a bound above one fixed cutoff to a bound valid at every threshold V is not proved. Assuming the ladder only strengthens the no-go (absorption holds even with the ladder in force), and Theorem 7.4 needs no ladder at all; nothing in this paper depends on that bridge.

- **Heuristic pricing.** For classes without an exhibited witness (the nonzero rows of the residual table; the claim that no undiscovered configuration class carries a strictly positive marginal value), the evidence is converged column-generation pricing ($S = 200$ restart protocol), which is heuristic adversary-optimality; it is never load-bearing for the exact-zero claims, which are witness-backed LP equalities.

- **Pair-channel residue** (Section 4.3 and the companion note cited there): (O1) The crowding cap's ledger constant is interval-certified to $w \leq 0.98$ ($0.98465 < 1$); the full-family ratio 0.9775 is a grid artifact — on $w \in (0.98, 1]$ the ledger fails (certified > 1) and coverage is adversarial numerics (+17.46, re-run at the sliver) plus the 8/9 backstop; the cap itself on $w \in (0.82, 0.98]$ remains granted-not-proved (its $w \leq 0.82$ self-consistency is interval-certified on both sides). (O2) Single-pair depths in $(0.45, 1.0)$ at unrestricted mass, multi-pair columns at the deep-probe depths $w \in \{1.5, 2\}$, and multi-pair configurations outside the admissible class (depth beyond $w = 2$, or unbounded mass), are covered by numerics plus the 8/9 backstop only — not needed for any LP statement. (O3) The sharp conjecture (Section 4.3) is open; near-tight configurations (+0.03) show that the constant is not easily improved. (O4) The certified capacity constants are at $N = 64$; the $N = 128$ re-run reproduces every identity and inequality with $\leq \sim$2.5% drift in the headline certified constants (deep-tail table values, $y \geq 1.5$, drift up to ∼8%), moves the deep-pair threshold of Theorem 4.6 from $d \geq 1.0$ to $d \geq 1.1$, and shows the near-endpoint ledger failure at $N = 128$ as well (smaller sliver, better capped constant 0.9633). (O5) All pair-channel constants are for the flat $\lambda = 1$ window; none transfer to the MT-cosine window without recomputation — consistent with that window's residual row above.

- **The formalized ceilings.** The 0.6818287 bandwidth-one ceiling belongs to the window-optimized certificate class; the 0.8453 trace/Frobenius cap stands on $\kappa \geq 2/\sqrt{3}$. The flat/flat row system used here consumes neither, and neither is revisited here: the claim made here is that cubic augmentation moves nothing at the flat/flat operating point and produces only the bounded residuals above elsewhere.

8.4 Outlook: the λ' payoff curve (outside the proven ladder)

The range $\lambda' > 1/2 + o(1)$ lies outside the proven $\vartheta < 1$ ladder regime; the entries below are payoff-curve data for the moderate-deviation target $\mathrm{MD}(\lambda, \eta)$ — the open problem of upgrading the fixed-λ divergent-cutoff spectral-tail statement, available for every $\lambda \in (1/2, 2/3)$ only above $V_0 = T^{2\lambda-1+\varepsilon}$, to an all-$V$ count ladder $n(V) \leq C_{\mathrm{led}} N V^{-4}$ valid down to $V \geq T^{\eta}$ — not certificate claims, and column generation ran at reduced budget, so the positive values are heuristic and upper-anchored.

λ'	ε	P_{base}	P_{full}	δ_0
0.55	0.05	0.8053310433508489	0.8053310433508484	0
0.55	0.02	0.8251787278264935	0.8251787278264899	0
0.60	0.05	0.8058514020011891	0.8062187975519491	$+3.67 \times 10^{-4}$
0.60	0.02	0.8256842190868753	0.8259625547887860	$+2.78 \times 10^{-4}$
0.65	0.05	0.8059503048717801	0.8086104956189072	$+2.66 \times 10^{-3}$
0.65	0.02	0.8263688470852664	0.8278795159840451	$+1.51 \times 10^{-3}$

The cubic row's marginal value turns positive between $\lambda' = 0.55$ and 0.60 — precisely where the flat-window moment margin $\mathrm{Marg}(\lambda) = 2m_2 - m_3$ changes sign (Corollary 3.6: $\lambda_0 = 0.610511$) — and grows toward the Rudnick–Sarnak endpoint $\lambda' = 2/3$. The first configuration that could pay therefore sits at $\lambda' \gtrsim 0.6$; consuming it would require extending the proven large-value ladder into exactly the moderate-deviation regime $\mathrm{MD}(\lambda, \eta)$, which is left open here as a community target. At the proven operating point $\lambda' = 1/2 + o(1)$, the value is exactly zero, which is what this paper proves; the curve above marks the boundary of that statement.

9 Consequences

1. **`lemmaR_tight` stands strengthened.** The two-moment degeneracy is not broken; it now carries an exact-zero certificate: two-bandwidth flat-window augmentation — λ'-Frobenius, signed cubic, all-V ladder, capacity fuzz, at $\lambda' = 1/2$ — adds exactly zero to the two-moment optimum at $N = 64/128$ scale, for both benchmarks, including inside the near-CUE class (by slack). This upgrades the standing conclusion from "the two-moment certificate is exhausted"

to "exhausted, and robust under Rudnick–Sarnak-range cubic augmentation with capacity control."

2. **The V-dependent bridge is not warranted by the cubic payoff alone.** What that bridge would buy for the cubic block is $\delta_0 = 0$. The all-V spectral-tail statement keeps its standalone value as the first such statement for the Weil–Gabor matrix, and its other consumers are unaffected.

3. **What is left standing, and what remains open, along this line:** (i) the all-V spectral-tail statement as a standalone analytic result (all-V count ladder with explicit C_{led}, $\vartheta < 1$ strict, C^3 taper); separately, its fixed-λ divergent-cutoff form, $V_0 = T^{2\lambda-1+\varepsilon}$, extends to all $\lambda \in (1/2, 2/3)$ via the fourth-moment Montgomery–Vaughan branch alone — a divergent-cutoff statement, subject to Theorem 7.4's vacuity, not an all-V ladder; (ii) partial orthonormalization off the $o(N)$-dimensional exceptional subspace, where a divergent cutoff suffices (unlike the cubic LP) — rank-perturbation bookkeeping already in the formalized linear-algebra library; (iii) the near-critical observation $X^2/T = (\log T)^{2\vartheta} =$ polylog, shrinking the shift range needed by quartic Hardy–Littlewood-type input for shifted prime pairs to logarithmic — a possible route via averaged Goldston–Montgomery / Montgomery–Soundararajan results; (iv) $\mathrm{MD}(\lambda, \eta)$ as a community target, now carrying the quantified payoff curve of Section 8.4.

4. **Methodological consequence.** The instrument is reusable independently of the answer it returned here: an explicit-configuration LP whose configurations are realizable by construction, clustered null budgets, an explicitly fixed decision rule and stopping criterion, absorption claims backed by an exhibited witness or not made at all, and the robustness checks of Section 4.4. Theorem 3.4 shows what the structural half of that instrument buys: it refutes the "+8 vs 0" doubles-versus-pairs separation premise outright, with no LP involved.

10 Formalization: what is machine-checked, and what remains

The core of Sections 3 and 4 is formalized, compiled, and machine-checked in Lean 4 against Mathlib, with no `sorry`, no `native_decide`, and no numeric certificates. Three files under `lean/Zeta23/PairCeiling/` of the accompanying repository carry it (Lean 4.33.0-rc2; Mathlib rev `51e6992`; the build completes successfully):

- `GridParseval.lean`. `flat_band_trace_sq` — Lemma 3.8, proved over any integral domain carrying a primitive M-th root of unity, for any band of M consecutive harmonics; and `trace_sq_grid` — Theorem 3.9, the literal $\operatorname{tr} \widehat{G}^2 = \sum_k m_k^2$ collapse on the grid. Supporting: character orthogonality on $\mathbf{Z}/M$, DFT Parseval in the integer-mark pairing, the band reindexing, and the triangular-weight telescoping.

- `GridWitness.lean`. `vacancy_F1` — the $N = 64$ vacancy lattice has $F_1 = \sum m^2 = 64$ for *every* vacancy position; `vacancy_half_band_value` — its $\lambda' = 1/2$ half-band row is exactly $F' = 4128/33$, again for every vacancy position, the hand-checkable worked value of Section 3.4 and the witness column, its integer core 136224 checked by `decide +kernel` (kernel-checked, not `native_decide`); and `half_band_alive` — $F' = 4128/33 \neq 64 = \sum m^2$, which is the "alive" half of Proposition 3.10, i.e. the two-bandwidth decoupling itself, in witness form. The finite/algebraic half of Proposition 3.10's display is formalized as `sum_band_pair_tri` / `half_band_fold`, together with `sum_window_residues` — that the s-window is a complete residue system mod 65, so no residue-class telescoping can occur, unlike the $\lambda = 1$ band.

- `GridCorner.lean.` `mark_one_count_ge` — Lemma 4.2 at the simple-fraction level; `grid_corner_pointwise` — Theorem 4.3 pointwise on the grid class; `grid_corner_law` — the same in law form, i.e. the LP lower bound with no discretization; and `grid_corner_attained` — exact attainment at $N = 64$, $\varepsilon = 1/32$ by a marks-$\{1, 2\}$ column (12 doubles, 40 simples), meeting both corners with equality simultaneously.

Axiom footprint. `#print axioms` on each of the twelve theorems named above returns `[propext, Classical.choice, Quot.sound]` and nothing else — the three axioms of Lean's classical foundation. In particular there is no `sorryAx`, which any admitted step would introduce, and no `Lean.ofReduceBool`, which `native_decide` would introduce and which would put the compiler's evaluator into the trusted base. Full record, with the environment and the reproduction recipe: `results/a4-no-go/formalization-status.md` in the accompanying repository (see *Provenance, data and code*).

The formalization reaches this far for a structural reason: on the ψ_1-zero grid every datum is algebraic, and Theorem 3.9 turns the first step of the corner chain from an inequality into an equality ($F_1 = \sum m^2$ exactly), after which the two integrality levels finish in finite arithmetic.

What is not formalized, and what each would take.

- **Proposition 3.10's general position-sensitivity statement** — two grid configurations with the same mark multiset but different F'. Finite algebra at fixed N, but it needs a cyclotomic non-vanishing certificate; the precise gap is documented in `GridWitness.lean`'s header. The witness form is proved; the general form is not.

- **Theorem 4.3 off the grid** (atom-only, arbitrary positions) — needs the position-space form of Lemma 4.1, i.e. $F_1 \geq \sum m^2$ with the kernel $K = D^2$ psd. This is the one step the grid case gets for free.

- **Proposition 3.3 and Theorem 3.4** — polynomial identities; `ring`-adjacent, and the cheapest remaining items.

- **Theorem 3.5** — piecewise-polynomial integration against Mathlib's integral.

- **Theorem 4.9** — a finite Hermitian matrix, inertia subadditivity, one scalar inequality; formalization-friendly, and the natural next unit from the pair channel.

- **Theorem 7.1 and Theorem 7.4** — one page of real analysis (Tonelli plus a one-variable optimization) and a bookkeeping argument respectively.

- **The pair-channel capacity results of Section 4.3** — Python-certified, with interval arithmetic. Formalizing the certified curves is a larger undertaking than the rest combined.

- **The full LP witness with its dual certificate** — the exact-rational route is available: the witness law is three columns with rational weights, integer site subsets and marks in $\{1, 2\}$, every row value rational ($F_1 = 64, 96, 128$; F' and C' rational combinations of the triangular weights, e.g. 4128/33), so feasibility-plus-optimality is an exact-rational check in the LawN256/CeilingLaw256 architecture with one added row family. The near-CUE witnesses extend the same format — the pinning rows are rational box constraints on the stored form factors, and both witnesses already pass exact `Fraction` re-verification against the true $\tau_2 = 1$ box.

A The witness data

The primary witness (`results/a4-m2-gate/witness_N64.json`; cell: $N = 64$, matched budgets $B_1, B_2, B_3 = 84.71214548308222, 135.09544203420174, 304.5549106659001$, $\varepsilon = 0.05$, fuzz = none). All three columns live on the 65-site ψ_1-zero grid, sites $\theta_k = k\,(64/65)$, marks $\{1, 2\}$, no pairs. Site subsets, summarized (full integer site lists are recorded per column in the JSON, keys `law[].atoms` / `grid_sites`; the canonical description is the clean integer grid — the stored `clean_checks` record the maximum row-value deviation between the clean-grid and optimizer values of $F_1/F'/C'$, at most 4.1×10^{-5}, per column $3.4 \times 10^{-6}/1.6 \times 10^{-5}/4.1 \times 10^{-5}$):

- Column 1 (weight 0.5299906673474316): the vacancy lattice — 64 simples on 64 of the 65 sites (one vacancy). $N_d = 64$; $F_1 = 64$ exactly; $F' = 4128/33$ exactly (hand derivation: $4096/33 + 32/33$, Section 3.4); $C' = 245.4508724$.
- Column 2 (weight 0.16040139164388625): 16 doubles + 32 simples on an explicit 48-site subset. $N_d = 48$; $F_1 = 96$ exactly; $F' = 153.7604346$; $C' = 411.0865130$.
- Column 3 (weight 0.30960794100868216): 32 doubles on an explicit 32-site subset. $N_d = 32$; $F_1 = 128$ exactly; $F' = 135.1959163$; $C' = 301.3541285$ (clean-grid values; the stored law row has $C' = 301.3541693$).

Aggregates (computed from the stored law rows, which saturate the budget edges exactly): $E[F_1] = 88.94775275723633 = B_1(1 + \varepsilon)$; $E[F'] = 132.81813304702743 \in [128.34, 141.85]$; $E[C'] = 289.32716513260516 = B_3(1 - \varepsilon)$; $E[N_d] = 51.52612362142$, $E[N_d]/N = 0.8050956815846875$. Active rows: `F1_hi`, `Cp_lo` only. Exact `Fraction` re-verification: all 10 row checks true, objective deviation 5×10^{-13}.

Near-CUE witnesses (`results/a4-m2-gate/followup-near-cue.json`, keys `witness_N64`, `witness_N128`). $N = 64$: 64 columns, marks $\{1, 2\}$, no pairs, weights 6.2×10^{-5}–0.0753, N_d per column 50–60 (4–14 doubles on CUE-like positions); $E|c_j|^2 = j + 1$ at every $j = 1, \ldots, 63$ (all upper pinning rows active at $+\tau_2$); $E[N_d]/N = 53.32693798297409/64 = 0.8332334059839699$; only pinning rows (and mass) active. $N = 128$: 128 columns, same class, upper pinning edge at $j + 1 - 10^{-5}$ (interior shrink; `Fraction` re-verification against the true $\tau_2 = 1$ box passes, margin $+1.0 \times 10^{-5}$); $E[N_d]/N = 107.19000239208492/128 = 0.8374219$ (the witness was solved at the interior-shrunk box; its objective sits 4.0×10^{-8} above the Section 5.2 LP optimum 0.8374218534574983); $E[n_{\text{simple}}]/N = 0.6748$.

The three JSON files (`witness_N64.json`, `followup-near-cue.json`, `followup-p1.json`) are in the repository cited in *Provenance, data and code*, under `results/a4-m2-gate/`: each contains the full column data, law weights, per-row values, active-row lists, and the rational-verification records.

B Null-model Monte Carlo tables

Closed-form anchors (Theorem 3.5): $m_2(1) = 4/3$, $m_2(1/2) = 13/6 = 2.166667$, $m_3(1/2) = 5$.

Primary sampler (CUE eigenangles via QR-with-phase-fix of complex Ginibre, unfolded to circumference n; R adaptive to $\mathrm{SE}(m_3) \le 0.5\%$; batch-means SE with jackknife check):

All three extrapolations are consistent with the closed-form anchors.

n	R	$m_2(1)$	$m_2(1/2)$	$m_3(1/2)$
64	4000	1.32363(74)	2.11087(63)	4.75867(362)
128	2000	1.32867(77)	2.13845(71)	4.87769(411)
256	600	1.33115(66)	2.15231(63)	4.93758(373)
512	300	1.33199(86)	2.15958(67)	4.96843(413)
1024	120	1.33364(110)	2.16362(83)	4.98715(476)
2048	48	1.33400(94)	2.16505(83)	4.99323(504)
$1/n$ fit $\to \infty$		1.33394(63)	2.16682(52)	5.00056(311)
closed form		1.33333	2.16667	5

Independent sampler (independent QR sampler, fresh seeds), side by side at the two shared sizes — agreement within $\sim$1 combined σ per entry:

n	$m_2(1)$ indep. / primary	$m_2(1/2)$ indep. / primary	$m_3(1/2)$ indep. / primary
64	1.32317(105)/1.32363(74)	2.11120(84)/2.11087(63)	4.76099(491)/4.75867(362)
256	1.33045(132)/1.33115(66)	2.15305(103)/2.15231(63)	4.94059(616)/4.93758(373)

A third independent sampler (own seeds, alias-free frequency-matrix assembly) gave $m_2(1/2) = 2.11063(135)$ at $n = 64$ and $2.15273(100)$ at $n = 256$, $m_3(1/2) = 4.7589(79)$ and $4.9408(59)$ — each within $\sim 1\sigma$ of both prior samplers — with two-point $1/n$ extrapolation $m_2 \to 2.1668$, $m_3 \to 5.0014$. A third route for $m_3(1/2) = 5$ (real-space 2D determinantal quadrature) gave $4.9896 \to 4.9932$ under truncation refinement; a further independent quadrature gave $4.9924 \to 4.9960$.

Ladder and occupancy side data (matched $n = 64$ unless noted): null ladder floor $C_{\text{led}}^{\min} = 3.8188$ (3.9119 at $n = 128$); $E[n(V)]/n$ on $V = [2, 3, 4, \ldots] = [0.23868, 0.01364, 0, \ldots]$; $\max|\text{eig}|$ over 4000 draws $= 3.977$. Cross-term (occupancy ≥ 2) shares: Frobenius 0.52626, cubic 0.78986, against the analytic 54% (7/6 of 13/6) and 80% (4 of 5; of which the strictly two-point share is 3.5 and the three-point term T_3 the remaining 0.5).

Provenance, data and code

Method. The specification of the decision computation — objective, adversary class, constraint rows, decision rule and stopping criterion — was fixed in writing before the computation was run. That specification, the run record and the verification outputs are among the artifacts listed below.

Data and code availability. Every number in this paper comes from one of three places: the decision record, the verification suite written for this paper, or the Lean development. All three are version-controlled and are available in the accompanying repository `https://github.com/x67ai/riemann-rh-program`, at the tag `paper-2026-08-28`, which is the state this paper was prepared against. Paths below are relative to `rh-program/` in that repository. One exception: the working PDF corpus of fetched literature is not redistributed, for copyright reasons, and nothing in this paper depends on it — every external claim is cited to its published source. The components are:

- *The formalization record* (`results/a4-no-go/formalization-status.md`) behind Section 10: the environment (Lean 4.33.0-rc2, Mathlib rev `51e6992`), the build result, the theorem-by-theorem map to this paper's numbering, the `#print axioms` output for all twelve machine-checked theorems, and the reproduction recipe.

- *The specification of the corrected row system* (`results/adjudication-A4.json`): the garnish and squeeze computations, and the row system in its corrected form, against which the computation was run.

- *The specification of the decision computation* (`results/a4-m2-gate/SPEC.md`): kernels and units, null budgets and their closed forms with derivations, the adversary class and its realizability argument, the row system and capacity constant, the decision rules, and the flags.

- *The run report* (`results/a4-m2-gate/RUN-REPORT.md`): the outcome, the grid-decoupling mechanism, the sanity checks, every run and number produced, the witness, and every deviation from the specification.

- *The independent check* (`results/a4-m2-gate/AUDIT.md`): a Monte-Carlo check of the null budgets, an independent re-derivation of the budget identities and the block law, recomputation of the witness, LP re-solves over all 4003 columns, and eight robustness checks of the conclusion along independent axes.

- *The follow-up report* (`results/a4-m2-gate/FOLLOWUP-REPORT.md`): the near-CUE runs at $\tau_2 = 1$ for both N, the p_1 objective, and the riders attached to the use of [BGSTB24].

- *The raw data* (`results/a4-m2-gate/*.json`, `results/a4-m2-gate/runs/*.json`): the decision-LP result, the witness law at $N = 64$, the near-CUE and p_1 follow-up outputs, and the full scan records at $N = 64$ and $N = 128$ (including the tight-ε, near-CUE, window and λ'-exploration runs, and the $\lambda' = 1/2$ budget derivations), all as JSON.

- *The proof note* (`results/a4-no-go/theorems.md`): the six proofs incorporated as Sections 3, 4.1 and 7, with their verification scripts and JSON outputs (`results/a4-no-go/verify/`).

- *The pair-channel note* (`results/a4-no-go/pair-channel.md`): the pair-interference closure (Sections 4.2–4.3), including the proofs of Theorems 4.6 and 4.7, for which only the proof architecture is given here, with its independent verification suite `results/a4-no-go/verify/pairchan_*.py` and `pairchan_verify_out.json`.

- *The data digest* (`results/a4-no-go/data-tables.md`): the digest from which every number in this paper is cited, each row carrying its on-disk source (file plus JSON key or report section); its Section 10 records two numerical cautions observed here (the 940-record maximum is the $N = 128$ value 8.94×10^{-13}; the near-CUE level 0.833377/0.8333622/0.8332334 drift across dictionary snapshots is common-mode, the equality $\delta_0' = 0$ being snapshot-independent).

- *The proposal record* (`directions/A4-lindelof-lock.md`): the original proposal and the specification of the computation reported here.

References

[AF26] L. Alpöge and R. Furman, *More than two thirds of the zeta zeros are simple and on the critical line*, arXiv:2608.13637 (v1, 13 Aug 2026; v2, 19 Aug 2026 — the arXiv record carries these two versions and no others), with its sorry-free Lean 4 formalization (Theorems A–E; `lemmaR_tight` in `Zeta23/ZeroSide/TightMult.lean`; the PairCeiling library, LawN256/CeilingLaw256; Theorem 5.8/Remark 5.10 prime-side moments). Two separately circulated revisions, neither of them an arXiv version, differ in their sectioning, and this paper cites them separately: (i) the 35-page version, *More than two thirds of the zeros of the Riemann zeta function lie on the critical line*, whose §7.5 "Limits of the method" carries the item (e) quoted in Section 1.2; and (ii) the 17-page version, called v5 here, *... are simple and on the critical line*, in which the same material appears, in a shorter form, as §7.2(e), §7.2 there being titled "Sharpness and limits of the method". Where this paper writes §7.5(e) it means (i); where it writes §7.2(e) it means (ii). Both readings are given in Section 1.2.

[BGSTB24] S. A. C. Baluyot, D. A. Goldston, A. I. Suriajaya, C. L. Turnage-Butterbaugh, *An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta function*, arXiv:2306.04799 (v1, 7 Jun 2023); Acta Arith. **214** (2024), 357–376; doi:10.4064/aa230612-20-3. Cited for: Theorem 1 (unconditional pointwise $F(\alpha)$ on the closed band $0 \leq \alpha \leq 1$, depth-weighted, multiplicity-counted), the Lemma 5 kernel class, and the Remark after Theorem 2 (rider 1). See also the same authors' sequel on the horizontal distribution of zeros, *Pair correlation of zeros of the Riemann zeta function I: proportions of simple zeros and critical zeros*, arXiv:2501.14545 (v1, 24 Jan 2025; v2, 21 Nov 2025; unpublished as of August 2026), which works under a narrow-vertical-box hypothesis.

[LR20] J. C. Lagarias and B. Rodgers, *Higher correlations and the Alternative Hypothesis*, Q. J. Math. **71** (2020), no. 1, 257–280; doi:10.1093/qmathj/haz043; arXiv:1905.12123. Cited in Section 1.5: they prove that the currently known band-limited n-level correlation data for zeta zeros — Montgomery ($n = 2$), Hejhal ($n = 3$), Rudnick–Sarnak ($n \geq 4$) — does not rule out the Alternative Hypothesis, by exhibiting a point process on $(1/2)\mathbf{Z}$ matching the sine process against all such test functions. Tao obtained the same conclusion independently, by different methods (cited there).

[LR21] J. C. Lagarias and B. Rodgers, *Band-limited mimicry of point processes by point processes supported on a lattice*, Ann. Appl. Probab. **31** (2021), no. 1, 351–376; doi:10.1214/20-AAP1592; arXiv:1907.03391. Their Theorem 1.4 and Proposition 3.4 show that above the Nyquist bandwidth the mimicking correlations are unique, while at critical sampling they are not — the same degeneracy this paper's Theorem 3.9 exploits, in the same place; the consequences for the scope of the absorption are discussed at the point of citation in Section 3.4.

[Mon73] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in Analytic Number Theory (St. Louis, 1972), Proc. Sympos. Pure Math. 24, Amer. Math. Soc., 1973, 181–193. The source of the extremal values 5/6 and 2/3 whose bandwidth-one analogs appear in Theorem 4.3 and Corollary 4.4.

[Hej94] D. A. Hejhal, *On the triple correlation of zeros of the zeta function*, Internat. Math. Res. Notices **1994**, no. 7, 293–302; doi:10.1155/S1073792894000334. The $n = 3$ correlation input named in Section 1.2.

[RS96] Z. Rudnick and P. Sarnak, *Zeros of principal L-functions and random matrix theory*, Duke Math. J. **81** (1996), no. 2, 269–322; doi:10.1215/S0012-7094-96-08115-6. The n-level range that names this paper's operating window. Scope, relevant to the cubic budget center (Section 8.3): their unconditional n-level theorem is for smooth test functions; taking a characteristic function requires RH. Section 3.3's determinantal ρ_3 and the $c_3(\lambda')$ identification invoke the smooth statement.

[CGG98] J. B. Conrey, A. Ghosh and S. M. Gonek, *Simple zeros of the Riemann zeta-function*, Proc. London Math. Soc. (3) **76** (1998), 497–522; doi:10.1112/S0024611598000306. The origin of the second integrality level that Lemma 4.2 and Theorem 4.3 consume, made unconditional in Alpöge and Furman's §7.2(c).

[BHB13] H. M. Bui and D. R. Heath-Brown, *On simple zeros of the Riemann zeta-function*, Bull. London Math. Soc. **45** (2013), no. 5, 953–961; doi:10.1112/blms/bdt026; arXiv:1302.5018. The conditional simple-zero input quoted in Section 1.2; their unconditional-on-RH distinct-zeros constant, as published, is 0.84665.

[CGdL20] A. Chirre, F. Gonçalves and D. de Laat, *Pair correlation estimates for the zeros of the zeta function via semidefinite programming*, Adv. Math. **361** (2020), 106926; doi:10.1016/j.aim.2019.106926; arXiv:1810.08843v2. The semidefinite-programming line for pair-correlation extremal problems, of which this paper's LP is the bandwidth-one marked-configuration analog. Two specific contacts. (i) Their inequality (7), $2N_s(T) \leq \sum_\rho (m_\rho - 2)(m_\rho - 3)/m_\rho = N^*(T) - 5N(T) + 6N_d(T)$, credited there to an argument of Ghosh, is the published instance of the same per-multiplicity integrality device Lemma 4.2 uses, at the reciprocal-weighted third level $(m-2)(m-3) \geq 0$; Lemma 4.2(a) is the unweighted second level $(m-1)(m-2) \geq 0$. (Their semidefinite programming is applied to the auxiliary-function class of their Section 3, not to (7); (7) is elementary integer arithmetic.) (ii) Their Corollary 3 is the corresponding published record for distinct zeros: $N_d(T) \geq (0.8477 + o(1))N(T)$ on RH and $(0.8486 + o(1))N(T)$ on GRH.

[GdLL25] F. Gonçalves, D. de Laat and N. Leijenhorst, *Multiplicity of nontrivial zeros of primitive L-functions via higher-level correlations*, Math. Comp. **94** (2025), no. 354, 2041–2058; doi:10.1090/mcom/4005; arXiv:2303.01095. The closest published third-party work to this paper: the same higher-level-correlation-versus-multiplicity question, attacked by semidefinite programming rather than by a marked-configuration LP. Cited in Sections 1.2 and 3.2. Their equation (5), $N_n(T) = \sum_k S(n,k)M_k(T)$ with $S(n,k)$ the Stirling numbers of the second kind, is the change of basis whose $n = 3$ case is the atom half of Theorem 3.4 here.

[FGL14] D. W. Farmer, S. M. Gonek and Y. Lee, *Pair correlation of the zeros of the derivative of the Riemann ξ-function*, J. London Math. Soc. (2) **90** (2014), no. 1, 241–269; doi:10.1112/jlms/jdu026. Cited only in Corollary 4.4, to disambiguate a numerical coincidence: their published RH-conditional lower bound for the proportion of distinct zeros is $N_d(T) > 0.8051\, N(T)$ (their Corollary 1.4). The two-author 2008 preprint arXiv:0803.0425 is an earlier version whose Corollary 1.4 gives 0.6544; the constant 0.8051 is the published version's.

[KLS07] T. Kuna, J. L. Lebowitz and E. R. Speer, *Realizability of point processes*, J. Stat. Phys. **129** (2007), no. 3, 417–439; doi:10.1007/s10955-007-9393-y; arXiv:math-ph/0612075. The truncated-moment/realizability problem that Section 2.4's adversary-realizability argument re-derives in the finite-circle setting.

[KLS11] T. Kuna, J. L. Lebowitz and E. R. Speer, *Necessary and sufficient conditions for realizability of point processes*, Ann. Appl. Probab. **21** (2011), no. 4, 1253–1281; doi:10.1214/10-AAP703; arXiv:0910.1710. The sequel, which settles the general finite-order truncated-moment realizability question; Section 2.4's finite-circle argument is a special case in their setting rather than a new problem.

[CdLS22] H. Cohn, D. de Laat and A. Salmon, *Three-point bounds for sphere packing*, arXiv:2206.15373 (v1, 30 Jun 2022; unpublished as of August 2026). The comparandum: in the sphere-packing analog the third-order augmentation *does* pay, improving on the two-point Cohn–Elkies bound in several dimensions; here it is worth exactly nothing.

[KS66] S. Karlin and W. J. Studden, *Tchebycheff Systems: With Applications in Analysis and Statistics*, Pure and Applied Mathematics, Vol. XV, Interscience Publishers (John Wiley & Sons), New York, 1966. The classical Chebyshev–Markov–Stieltjes/Christoffel background for the fact that odd moments do not improve one-sided bounds — the general form of the cubic blindness of Theorem 3.4.